



**ACL-FDM: FREQUENCY DIVISION
MULTIPLEXING
TRANSMITTER AND RECEIVER
TRAINER KIT
EXPERIMENTAL MANUAL**

.... A MESSAGE FROM



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SAFETY RULES

Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook with you for any further help.

After unpacking, set all accessories in order so that they are not lost and check the kit integrity with respect to the packing list. In particular, check that the kit is integral and shows no visible damage.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

In **ACL-FDM: FDM TRANSMITTER AND RECEIVER TRAINER KIT** modules, signals are to be monitored with an oscilloscope as explained in the manual. The scope input channel should be ac coupled, unless otherwise indicated. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180 Ω resistance in series with normal oscilloscope probe.

This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. If any failure resulting from a defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the event warranty service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

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FEATURES

- Onboard clock generator
- Onboard sine and carrier wave generator
- 2 channel FDM communication system
- DSB-SC AM modulator
- DSB-SC AM demodulator
- Band Pass filters
- Low Pass filters
- Out of band signal transmission
- Voice communication (optional)
- Switch faults:
8 switch faults on ACL-FDM are provided onboard to study different effects on the circuit
- Interconnections facilities:
Sockets and connecting cords are provided for connections onboard and the external boards when required.
- Test points:
All relevant test points are brought-out for observation Observations are carried out on an oscilloscope

TECHNICAL SPECIFICATIONS

CH1

Input Frequency Range: 500 Hz to 1 KHz

Amplitude : 0 to 2Vpp

CH2

Input Frequency Range: 1 KHz to 2 KHz

Amplitude : 0 to 2Vpp

Carrier Generator

Sub Carrier : 1 KHz to 20 KHz and 1 KHz to 30 KHz with
Amplitude of 2Vp-p

Pilot Carrier : 256 KHz

Transmitter

High frequency transmission using DSB

Multiplexing : Frequency Division Multiplexing

Channel : 02 Nos.

Band Pass Filter

Pass Band Filter 1 & 3

Frequency range : 7 to 11 KHz with $f_c=9$ KHz

Filter 2 & 4

Frequency range : 18 to 22 KHz with $f_c=20$ KHz

AM/ DSB Modulator

Modulation: Amplitude Modulation Double Side Band with suppressed carrier using
Balanced Modulators

Demodulation: Amplitude Demodulation of DSB-SC using Balanced Demodulator

De-multiplexing : Frequency Division De-multiplexing

Low Pass Filter : 4th order butter-worth filter

Switch Faults: 8 Switch Faults are provided on board to study Different Effects on circuit

Inter-Connections: 2 mm Banana Socket

Test Points : 25 Test points are provided to observe various
Intermediate signals

Power Supply : GND, +5V, +12V, -12V

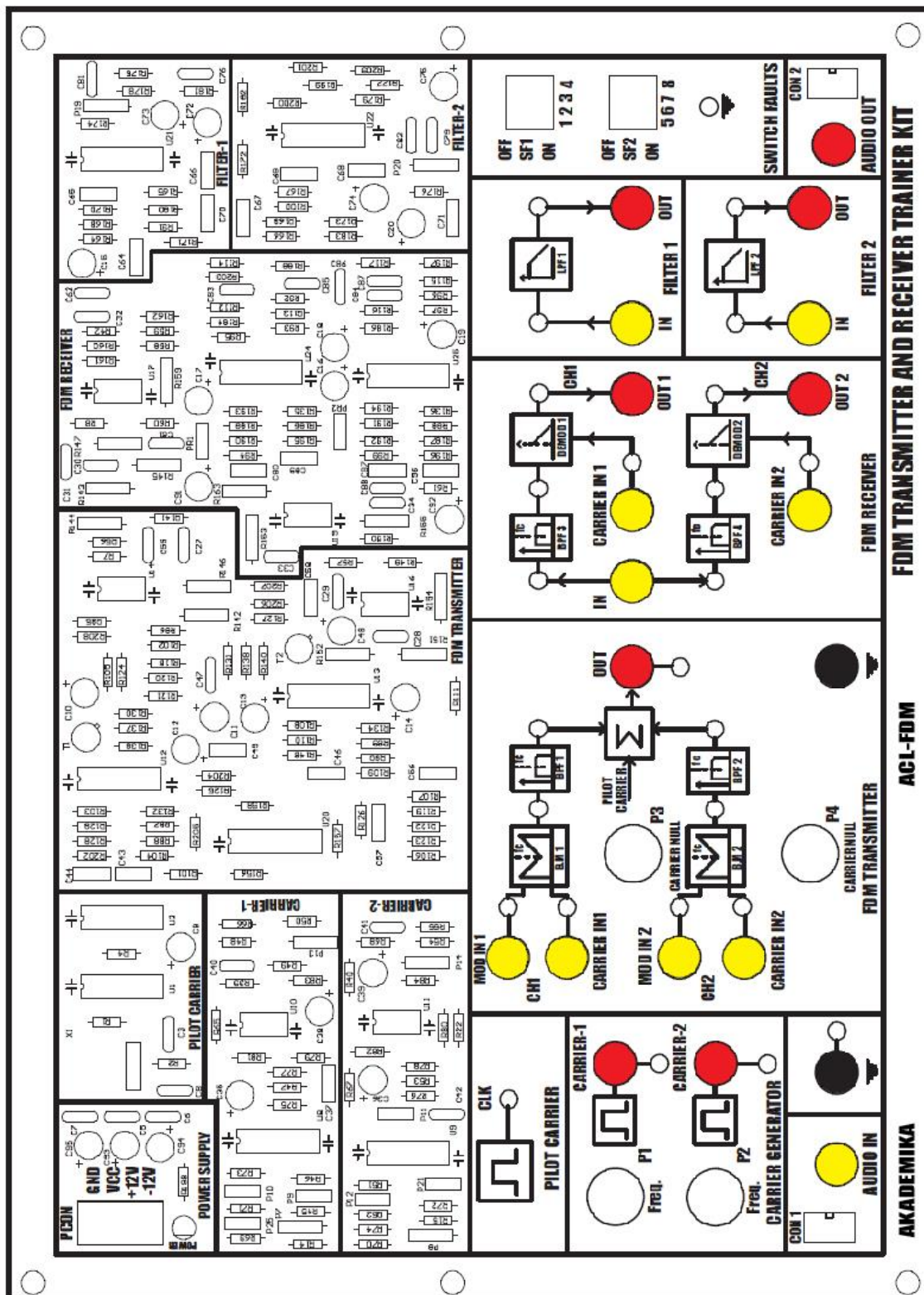
Optional : A I/O Audio Input Output Kit

Switch Faults : 8 switch faults

Accessories:

ACL-FDM			Quantity
1	Short Links PCS2 Patch cord	Male To Male	3
2	Mini Links PCS2 Patch cord	Male To Male	7
3	ACL-FDM EXPT Manual		1
4	Power Supply		1
5	Power cord		1

FUNCTIONAL BLOCK



INTRODUCTION

AKADEMIKA with the trainer, namely, “ACL-FDM” initiates the user to the study the Frequency Division Multiplexing Techniques using AM modulator, Band Pass Filters and Balanced Demodulators normally adopted in practice.

“ACL-FDM”: FDM TRANSMITTER AND RECEIVER TRAINER KIT

- **Pilot Carrier**

The 2.048 MHz crystal oscillators generate a 2.048 MHz clock. It is divided by ripple counter U2A (IC 74HCT393) to get the 256 KHz carrier frequency. 128 KHz clock is further divided by a ripple counter U2B (IC74HCT393) to get 64 KHz, 32 KHz, 26 KHz and 8 KHz clock frequencies, which is used for further application in the circuit.

- **Carrier Generator**

Carrier1

The integrated circuit U8 (ICL8038) generates sine (pin 2) and triangular waveform (pin 3). The capacitor, which is connected to pin 10, enables to select the frequency range (1 or 20 KHz max). The frequency is varied with the pot P7 FREQ. (P25 limits the inferior frequency). The trimmers P9 and P10 adjust the symmetry and the form of the sine signal. The sine is applied to the amplifier U10 (LF353) and then to the output. The frequency is varied using pot P1.

Carrier2

The integrated circuit U9 (ICL8038) generates sine (pin 2) and triangular waveform (pin 3). The capacitor, which is connected to pin 10, enables to select the frequency range (1 or 30 KHz max). The frequency is varied with the pot P8 FREQ. (P21 limits the inferior frequency). The trimmers P11 and P12 adjust the symmetry and the form of the sine signal.

The sine is applied to the amplifier U11 (LF353) and then to the output. The frequency is varied using pot P2.

- **FDM Transmitter**

Balanced Modulator 1 & 2

The integrated circuit LM1496 consists in a quadruple differential amplifier, driven by a further differential stage. The output signal consists in a constant, which multiplies the product of the two input signals CARRIER and SIGNAL. The trimmer NULL CARRIER enables the balancing or unbalancing of the circuit: in the first case the output is exactly the product of the input signals, and the circuit operates as balanced modulator; in the second case the output contains a fixed component of the signal CARRIER and the circuit operates as amplitude, which is taken from the transistor emitter.

Band Pass Filter 1

A band-pass filter has a pass band between two cutoff frequencies f_H and f_L , where $f_H > f_L$ and two stop-bands: $0 < f < f_L$ and $f > f_H$. The bandwidth of the band pass filter is therefore equal to $f_H - f_L$. Q is a measure of selectivity, meaning that higher the value of Q , the more selective is the filter or the narrower its bandwidth (BW). Q is also defined as the figure of

merit or quality factor of a filter. The relationship between Q , the 3-dB bandwidth and the center frequency f_c is given by

$$Q = f_c / BW = f_c / (f_H - f_L)$$

This is a narrow band pass filter ($Q > 10$), where the output voltage peaks at the center frequency. This filter uses only one op-amp. It has two feedback paths, hence the name multiple-feedback filter. The op-amp is used in the *inverting* mode. The filter is built around op-amp U14 (IC TL082) with pass band frequency range of 7 KHz to 12 KHz and cut off frequency of 9 KHz.

Band Pass Filter 2

The circuit is analogous to the one of the BAND PASS FILTER 1. This filter is built around op-amp U16 (IC LF356) with pass band frequency range of 18 KHz to 23 KHz and cut off frequency of 20 KHz.

Adder

Adder is formed using U20 (IC 4053). In this integrated chip, it will do summation of two signals applied to it. If two signals with different frequency components are applied to this chip we will get FDM output.

- **FDM Receiver**

Band Pass Filter 3

This is a wide band pass filter whose center frequency f_c can be defined as

$$f_c = (f_H f_L)^{1/2}$$

where f_H = high cutoff frequency (Hz)

f_L = low cutoff frequency of the wide band pass filter (Hz)

A wide band pass filter can be formed by simply cascading high pass and low pass sections and is generally the choice for simplicity of design and performance.

This filter is built around op-amp U17 (IC TL082) with pass band frequency range of 7KHz to 11KHz and cut off frequency of 9KHz.

Band Pass Filter 4

Refer to the circuit diagram as shown later in this manual sheet **6 of 9** of **ACL-FDM**. The circuit is analogous to the one of the BAND PASS FILTER 1. This filter is built around op-amp U19 (IC LF 356) with pass band frequency range of 19KHz to 22KHz and cut off frequency of 20KHz.

Demodulator 1

Refer to the electrical diagram sheet **7 of 9** of **ACL-FDM**, which includes a product detector U24 (LM1496); a low pass filter type RC across the output of the detector.

The product detector multiplies the two input signals between them and can be used alone, to demodulate the signals modulated in phase.

This demodulator multiplies signal coming from BPF3 with carrier-1 frequency resulting in demodulated output signal.

Demodulator 2

Demodulator which includes a product detector U25 (LM1496); a low pass filter type RC across the output of the detector.

The product detector multiplies the two input signals between them and can be used alone, to demodulate the signals modulated in phase.

This demodulator multiplies signal coming from BPF4 with carrier-2 frequency resulting in demodulated output signal.

Filter 1

Refer to the circuit diagram as shown later in this manual sheet **7 of 9** of **ACL-FDM**. Fourth order low pass butter-worth Filter is formed by U21 (IC TL084) using resistors and capacitors.

Filter 2

Refer to the circuit diagram as shown later in this manual sheet **9 of 9** of **ACL-FDM**. Fourth order low pass butter-worth Filter is formed by U22 (IC TL084) using resistors and capacitors.

Frequency Division Multiplexing

Multiplexing is a signal processing operation by which a number of independent messages can be combined into a composite signal suitable for transmission over common channel. The technique of separating these messages in frequency is referred to as frequency division multiplexing (FDM).

The principle of FDM is illustrated by Fig., where several input messages (three are shown) individually modulate the *subcarriers* f_{c1} , f_{c2} , etc., after passing through LPFs to limit the message bandwidths. We show the sub carrier modulation as SSB and it often is; but any of the CW modulation techniques could be employed, or a mixture of them. The modulated signals are then summed to produce the *baseband* signal, with spectrum $X_b(f)$, as shown in the Fig. (The designation "baseband" is used here to indicate that final carrier modulation has not yet taken place). The baseband time function $X_b(t)$ is left to your imagination.

Assuming that the sub carrier frequencies are properly chosen, the multiplexing operation has assigned a slot in the frequency domain for each of the individual messages in modulated form, hence the name frequency-division multiplexing. The baseband signal may then be transmitted directly or used to modulate a transmitted carrier of frequency f_c . We are not particularly concerned here with the nature of the final carrier modulation, since the baseband spectrum tells the story.

Message recovery or demodulation of FDM is accomplished in three steps portrayed by Fig. First, the carrier demodulator reproduces the baseband signal $X_b(t)$. Then the modulated sub carriers are separated by a bank of bandpass filters in parallel, following which the messages are individually detected.

The major practical problem of FDM is a cross talk, the unwanted coupling of one message to another. Intelligible cross talk (cross modulation) arises primarily because of non linearity's in the system, which causes one message signal to appear as modulation on another subcarrier, consequently, standard practice calls for negative feedback to minimize amplifier nonlinearity in FDM systems. (As a matter of historical fact, the FDM crosstalk problem was a primary motivation for their development of negative feedback amplifiers.)

Unintelligible cross talk may come from nonlinear effects or from imperfect spectral separation by the filter bank. To reduce the latter, the modulated message spectra are spaced out in frequency by guard bands into which the filter transition regions can be fitted. For example, the guard band between $X_{c1}(f)$ and $X_{c2}(f)$ in the given figure is of width $f_{c2} - (f_{c1} + W_1)$. The net baseband bandwidth is therefore the sum of the modulated message bandwidths plus the guard bands.

While the concept of FDM is quite simple, typical systems may be very elaborate. A case in point is the AT & T type L4 carrier system, in which 3600 voice channels (each with $W =$

4KHz nominally) are multiplexed together for transmission via coaxial cable. All modulation is single sideband, both USSB and LSSB, and the final baseband spectrum runs from 0.5 to 17.5MHz including pilot carriers and guard bands. All subcarriers are multiples of 4KHz synthesized from a common oscillator, and the 512KHz pilot provides synchronization. To avoid excessive guard-band width at the upper end of the baseband spectrum, the multiplexing is done by groups in four stages, an arrangement that also facilitates switching and routing of the various channels.

Figure depicts the first two stages of telephone multiplexing. The group multiplexer takes 12 input voice channels and puts them on subcarriers at 64,68,..., 108KHz using LSSB modulation. The resulting group spectrum is 48KHz wide, starting at 60KHz. Five such groups are then LSSB modulated on subcarriers at 420,468, 612KHz to produce a super group spectrum from 312 to 552KHz. Note that the suppressed subcarrier for group 1 would fall in the group 3 spectrum. Also note that the voice channel 1 in-group 1 one appears as a USSB signal at 354KHz. (Anyone who perchance is not fully convinced of the power of frequency domain analysis should consider the task of designing this system without the aid of spectral concepts.)

The given table lists the AT & T FDM hierarchy for reference purposes. The indicated frequency ranges are for the L4 system, but the channel groupings and bandwidths also apply to other systems. At any point in the hierarchy, a wideband video or digital data channel of comparable bandwidth may replace the specified number of voice channels.

Designation	Frequency range (L4 system)	Bandwidth	Number of voice channels
Group	60-108KHz	48KHz	12
Super group	312-552KHz	240KHz	60
Master group	564-3084KHz	2.52MHz	600
Jumbo group	0.5-17.5MHz	17MHz	3600

Frequency division multiplexing (FDM), unlike TDM, transmits and receives for the full 360 degrees of a sine wave. FDM used presently by the Navy may be divided into two categories. One category is used for voice communications and the other for teletypewriter system (tty) communications.

The normal voice speaking range is from 100 to 3,500 hertz. During single channel AM voice communications, the audio frequency amplitude modulates a single Rf (carrier frequency). However, in voice FDM, each voice frequency modulates a separate frequency lower than the carrier frequency (sub carrier frequency). If 3,500 hertz or more separates these sub carrier frequencies, they may be combined in a composite signal. This signal modulates the carrier frequency without causing excessive interference.

For e.g., the output of channel one is the voice frequency range of 100 to 3,500 hertz. The output of channel two is the combination of a different voice frequency with a sub carrier frequency of 4,000 hertz. The output of channel three is another voice frequency. This voice frequency combined with a sub carrier frequency of 8,000 hertz gives you an output frequency range of 8,100 to 11,500 hertz. The overall bandwidth for the composite modulation package shown is 100 to 15,500 hertz. Each separate channel occupies its own band of frequencies. The composite signal is used to modulate the carrier frequency of the transmitter.

Multichannel broadcast and ship/shore terminations use the FDM. With this system, each channel of the composite tone package of the broadcast is assigned an audio frequency. By multiplexing tty circuits, up to 16 circuits may be carried in any one of the 3000 Hz multiplexed channels described above. Don't confuse the two types of multiplexing. In the first case, 3000 Hz audio channels have been combined. In the second case, a number of dc the circuits are converted to tone keying and combined in a single 3000 Hz audio

channel. Figure 3-31 illustrates a 16-channel, tty-multiplexing system. The output of the dc pulsed circuits is converted to audio keying. Each channel has a separate audio center frequency. Channel frequencies range from 425 hertz for the lowest channel to 2,975 hertz for the highest channel. A mark in an individual tty loop keys an audio tone 42.5 hertz below the center frequency. A space in the input signal keys an audio tone 42.5 hertz above the center frequency. Let's look at an example. The mark and space frequencies for channel one are calculated as 382.5 hertz and 467.5 hertz, respectively (425 ± 42.5). Combining these keyed tones into a composite signal results in a tone package within a standard 3,000-hertz bandwidth. By occupying no more than 3,000 hertz of the audio spectrum, the output signal is suitable for transmission via radio or landline.

Amplitude Modulation

In Amplitude Modulation the amplitude of high frequency sine wave (carrier) is varied in accordance with the instantaneous value of the modulating signal.

Consider a sine signal $v_m(t)$ with frequency f

$$V_m(t) = B \cdot \sin(2\pi f \cdot t)$$

And another sine signal $v_c(t)$ is called modulating signal, the signal $v_c(t)$ is called carrier signal.

$$V_c(t) = A \cdot \sin(2\pi F \cdot t)$$

The signal $v_m(t)$ is called modulating signal; the signal $v_c(t)$ is called carrier signal.

Vary the amplitude of the carrier $v_c(t)$ adding the modulating signal $v_m(t)$ to A.

You obtain a signal $v_M(t)$ amplitude modulated, which can be expressed by:

$$v_M(t) = [A + k \cdot B \cdot \sin(2\pi f \cdot t)] \cdot \sin(2\pi F \cdot t) = A \cdot [1 + m \cdot \sin(2\pi f \cdot t)] \cdot \sin(2\pi F \cdot t)$$

With k = constant of proportionality.

Percentage modulation signal is defined as the value:

$$m = \frac{H-h}{A} \cdot 100 \quad K \cdot B$$

With reference to FIG.1.7, the modulation index m can be calculated in this way

$$m = \frac{H-h}{H} \cdot 100\%$$

Amplitude Modulation Generation

The circuits used to generate an amplitude modulation must vary the amplitude of a high frequency signal (carrier) as function of the amplitude of a low frequency signal (modulating signal).

In an AM transmitter we speak of:

High-level modulation, if the modulation is carried out directly in the last power stage, which is generally an amplifier in class C.

Low frequency modulation, when the modulation is carried out by stages, which are before the final power amplifier.

Semiconductor devices can be used, in case of low power or valve ones, when the required power is high.

Double-Sideband/Suppressed-Carrier

The amplitude-modulation technique called double-sideband/suppressed-carrier (DSB-SC) has an important advantage over regular AM (DSB-FC): The carrier is suppressed during the modulation process. As a result, most of the power in a regular AM transmission, which provides no information, is eliminated.

Example

Determine the power savings when the carrier is suppressed in a regular AM signal modulated to an index of 100%.

Solution

$$P_t = (1 + m^2/2) P_c, P_{sb} = P_c m^2/2.$$

The power saving is $(P_t - P_{sb})/P_t = 1/(1 + m^2/2) = 1/1.5 = 66.7\%$ for DSB-SC transmission.

The power savings of Example has its price, however. As will be obvious by simple inspection of the waveform of a DSB-SC signal, an AM rectifier cannot be used to demodulate DSB-SC. Demodulation can be achieved only if a locally generated carrier signal is introduced. It must not only have exactly the correct frequency (be frequency-coherent) but also have a phase very close to what the carrier would have if it had been transmitted; that is, DSB-SC demodulation must also be approximately phase-coherent.

The peak detector discussed for demodulation of regular AM (DSB-FC) will not yield the correct result for DSB-SC. For instance, when the input is the sinusoidal tone-modulated DSB-SC signal, the output of a peak detector will be the "cups" signal of Figure a.

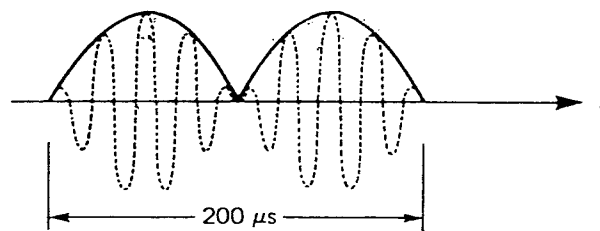
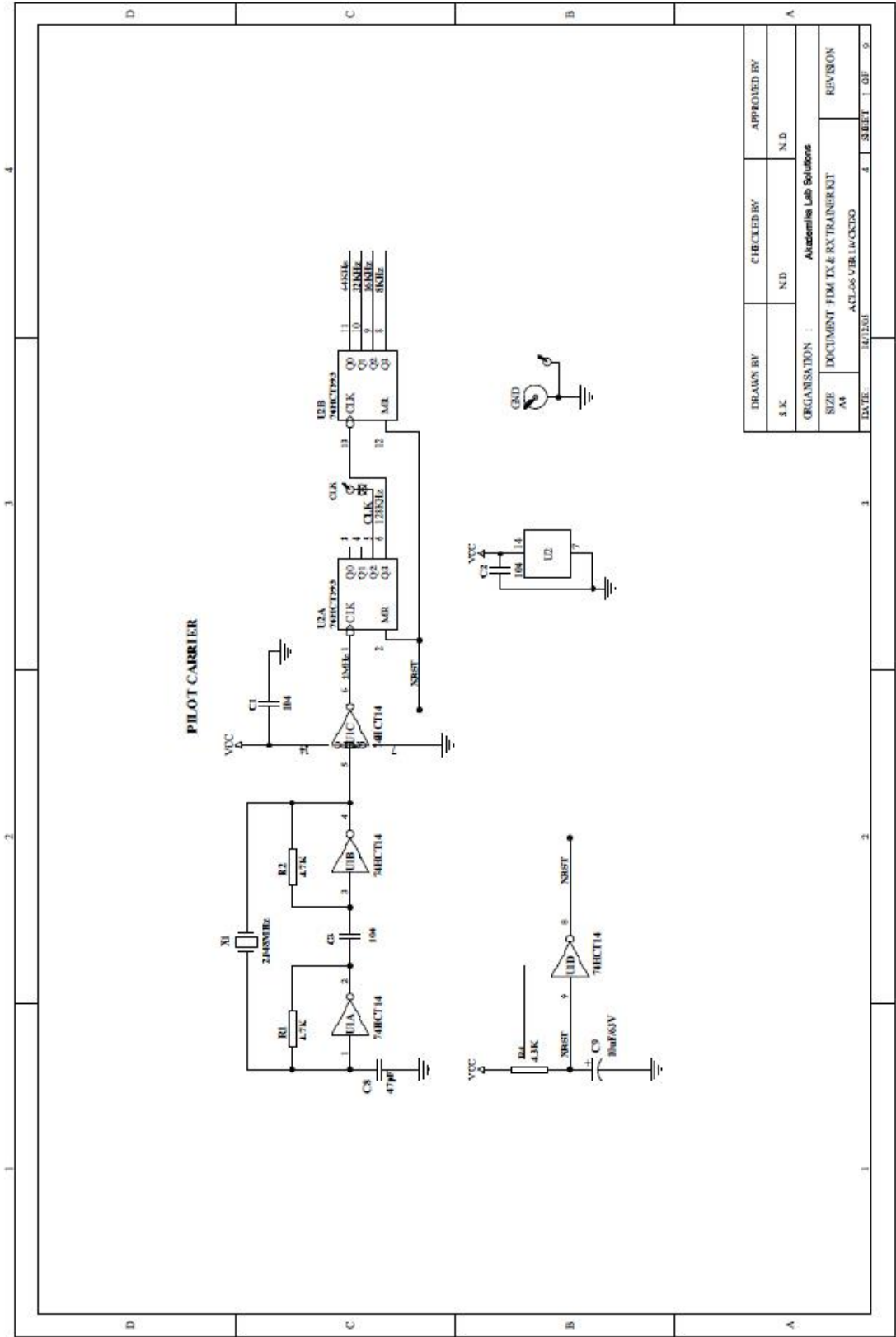


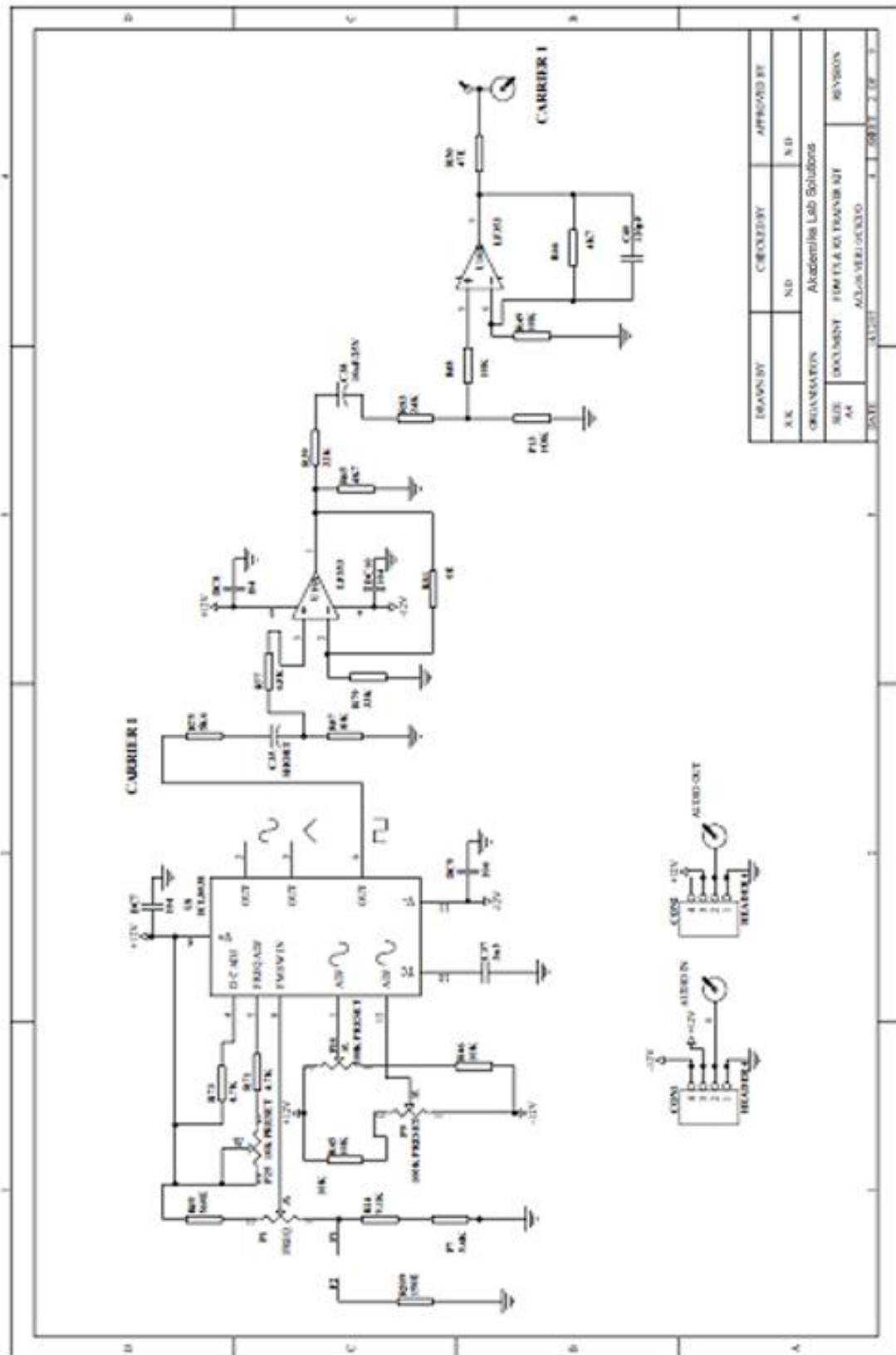
Figure a. Result of non-coherent demodulation of DSB-SC

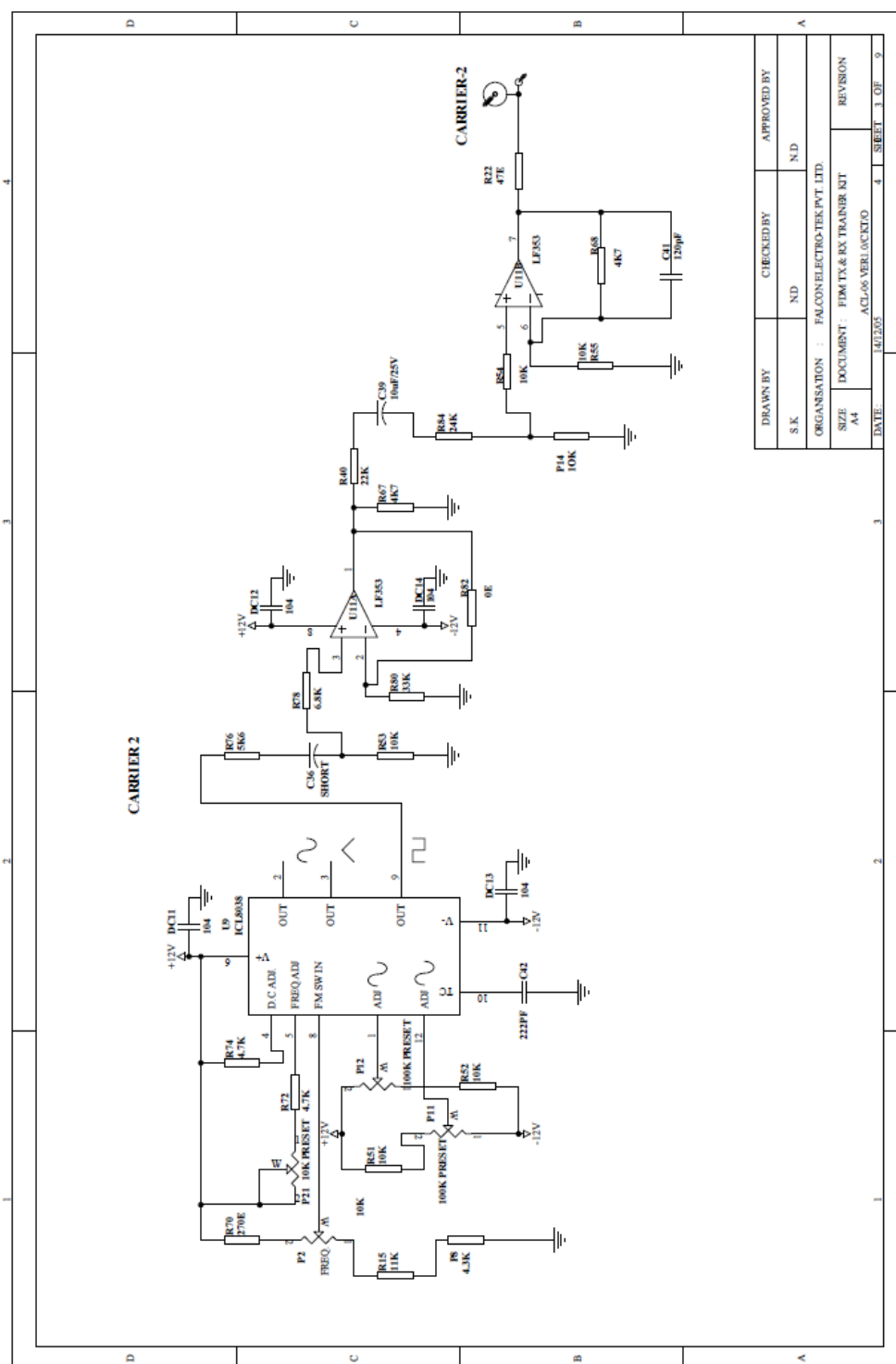
SWITCH FAULTS

- When we put the switch **1** of **SF1** in Switch Fault section to ON position, resistor R209 will get shorted to R14 in the CARRIER 1 section. This leads to change in carrier frequency i.e. center frequency of Balanced Modulator I and Band Pass Filter I causing instability in FDM Transmitter output.
- When we put the switch **2** of **SF1** in Switch Fault section to ON position, resistor R144 will get shorted to resistor R141 due to which center frequency of **BPF1** gets distorted. This leads to instability in **BPF1** output and further in FDM Transmitter output.
- When we put the switch **3** of **SF1** in Switch Fault section to ON position, it will short resistor R202 in parallel with pin no.5 of U12 in the BALANCED MODULATOR 1 section and ground signal. Due to this, biasing voltage of IC U12 gets affected results in disappearance of signal at BM1 output test point.
- When we put the switch **4** of **SF1** in Switch Fault section to ON position, it will short the resistor R203 in parallel with pin no. 5 of U24 (Demodulator 1) and ground. This will result in imbalance of bias voltage of IC U24 and distorted output at **OUT1** post of Demodulator 1 section.
- When we put the switch **5** of **SF2** in Switch Fault section to ON position, it will open resistor R134 in series from pin no. 5 of U13 in the BALANCED MODULATOR 2 section. It will result in imbalance of biasing voltage for IC U13 due to which output at **BM2** test point will disappear.
- When we put the switch **6** of **SF2** in Switch Fault section to ON position, it will open capacitor C32 from pin no. 5 IC U17 of the BPF3. Center frequency of BPF 3 gets distorted, which will result in unwanted signal at the output of **BPF3** section.
- When we put the switch **7** of **SF2** in Switch Fault section to ON position, it will open capacitor C64 with pin no. 3 of IC U21 of the FILTER 1 section. It will result in imbalance of filter 1 due to which saturated signal of sine wave will be observed at the OUT post of FILTER1
- When we put the switch **8** of **SF2** in Switch Fault section to ON position, it will affect the Product detector 2 section. Output is not obtained. Due to this, signal connected to pin no. 4 of U25 will be disconnected and it will result in distorted output at output test point of Demodulator 2.

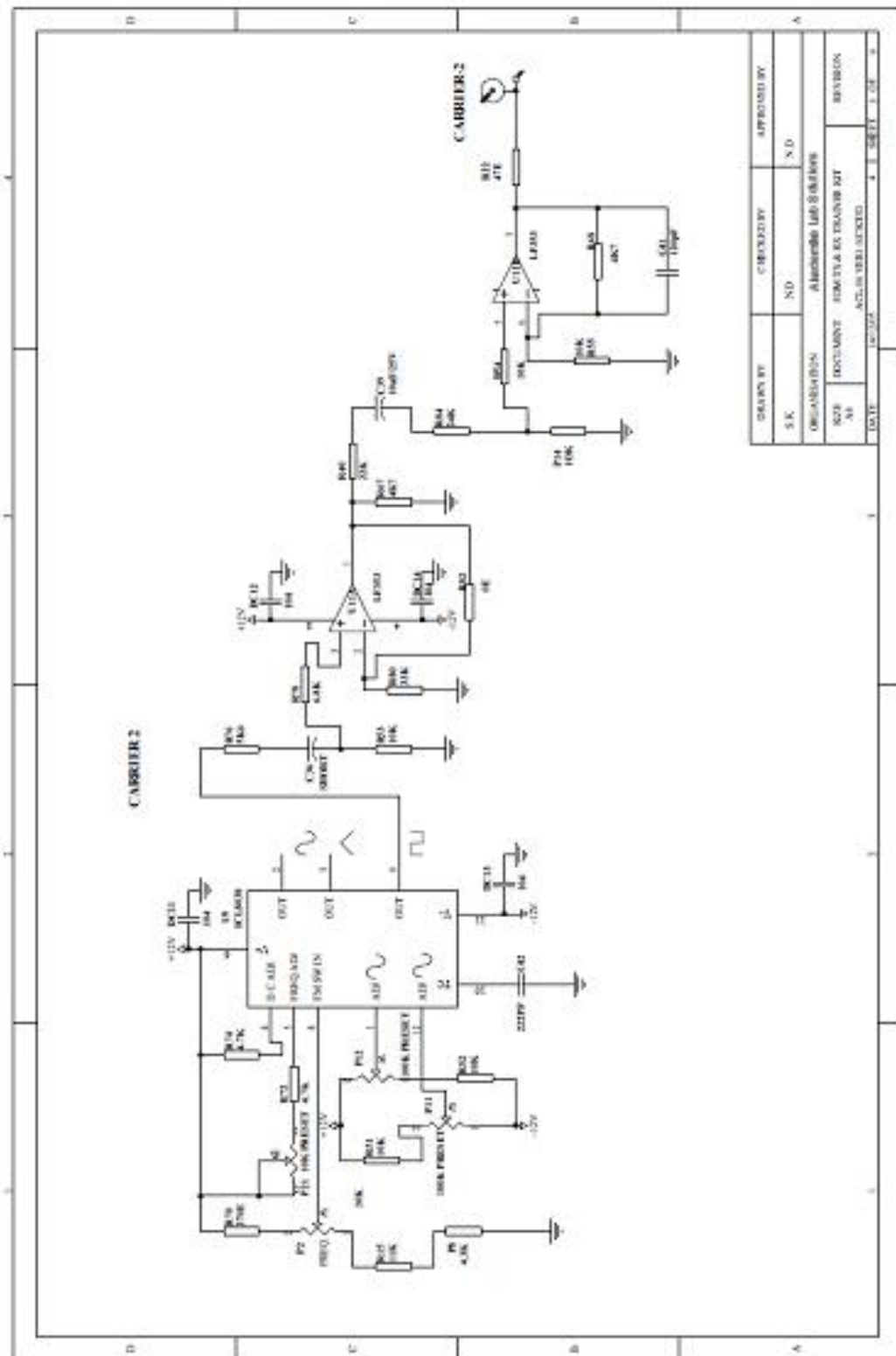
CIRCUIT DIAGRAM

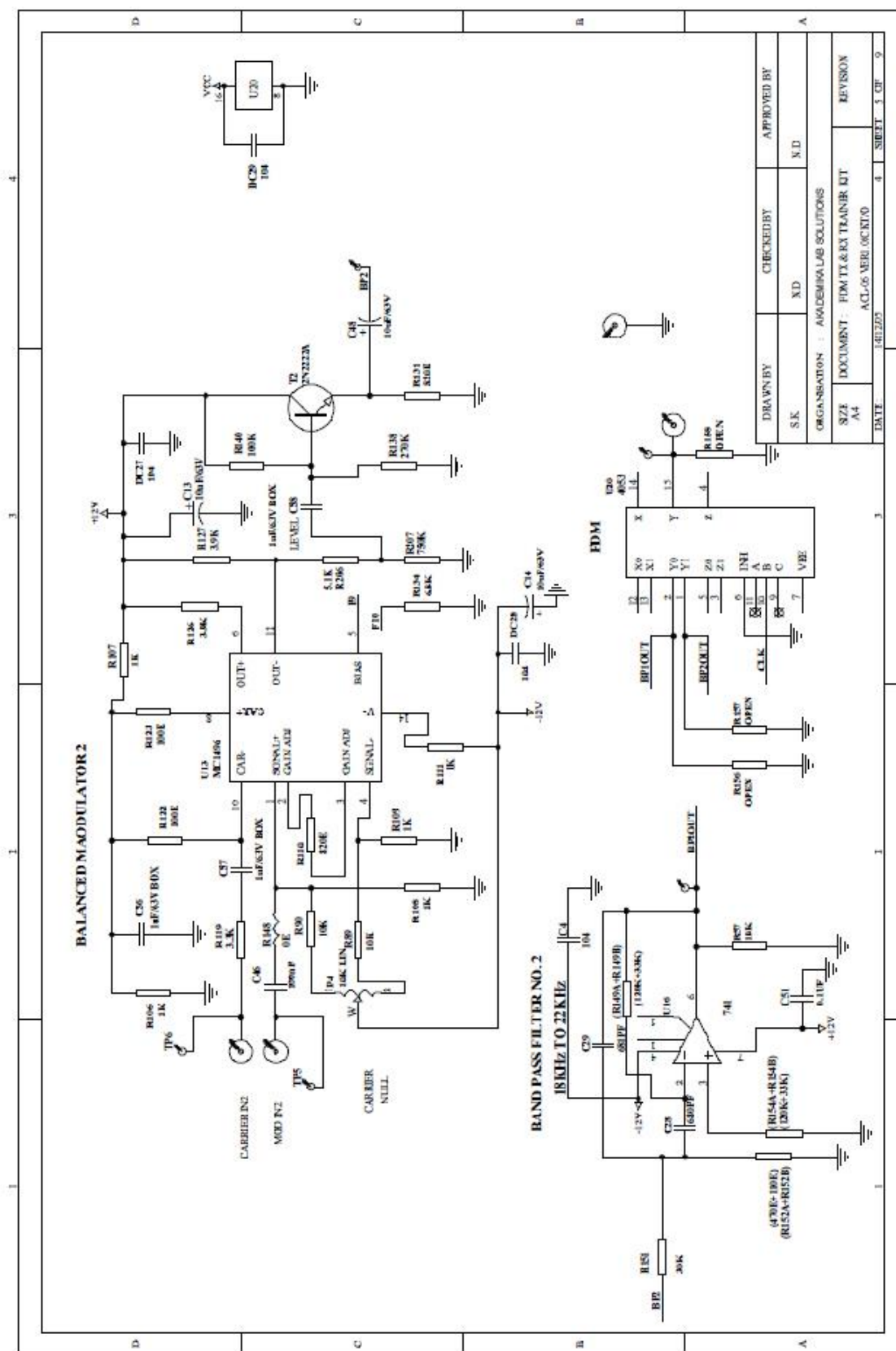


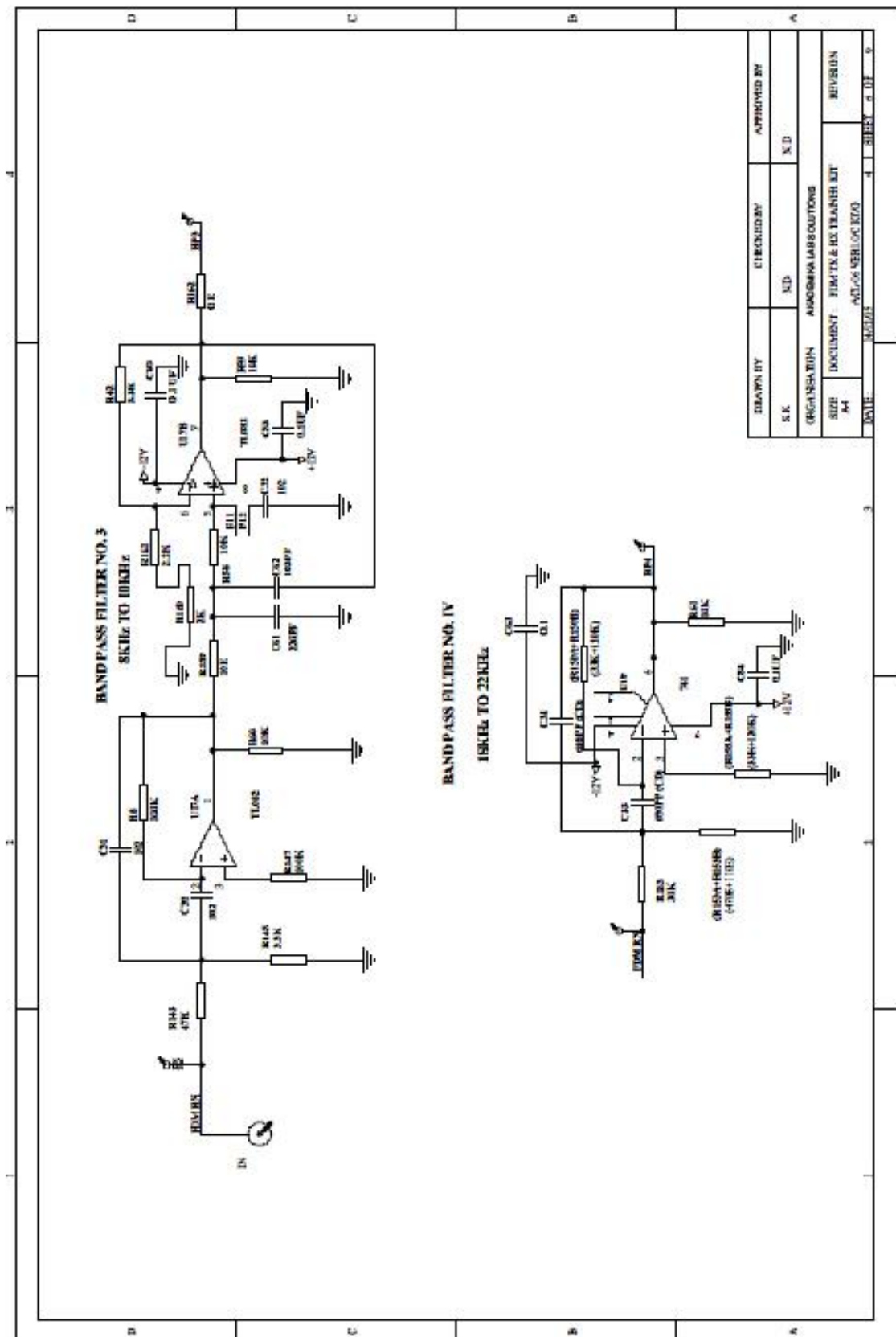


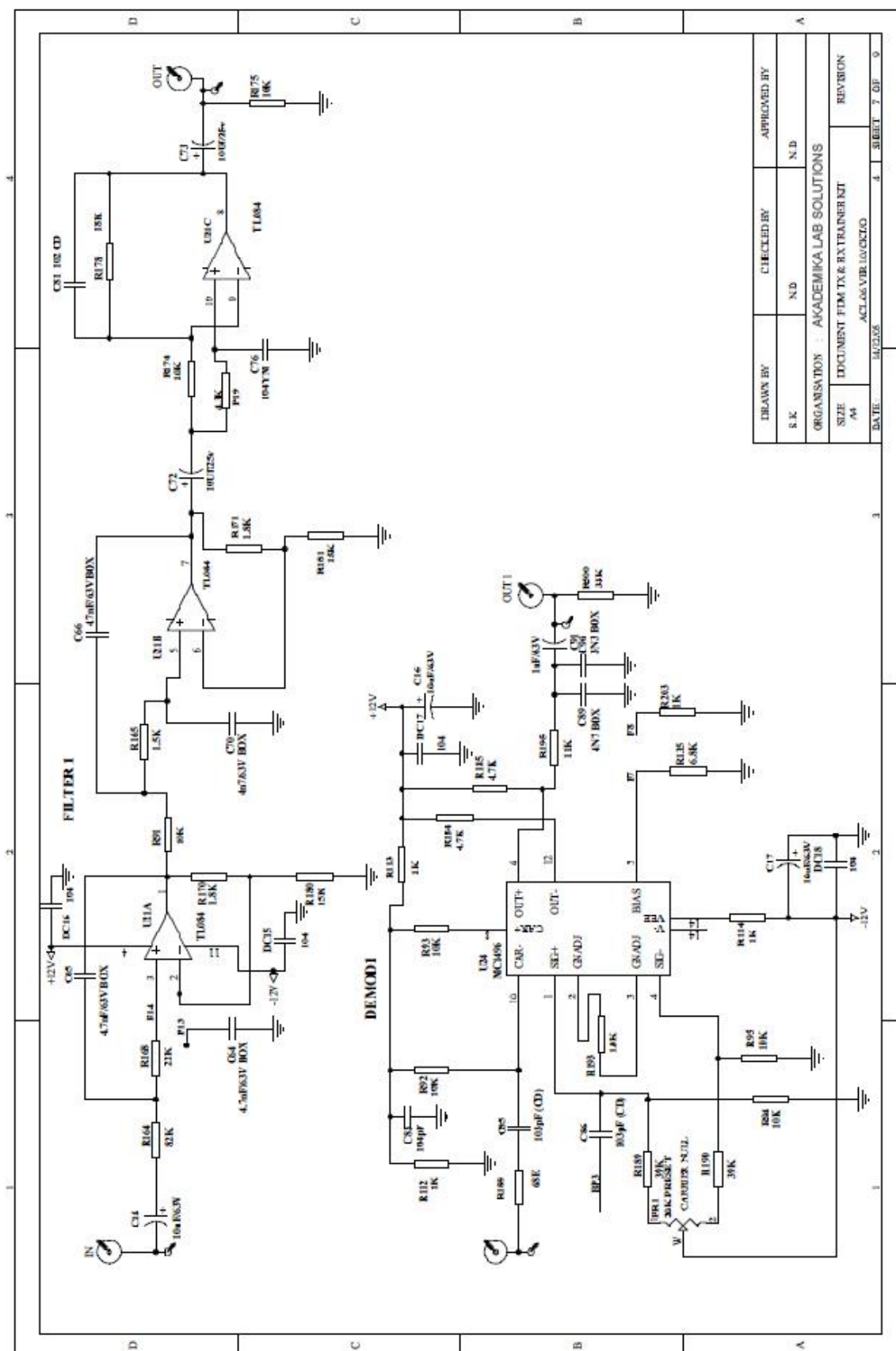


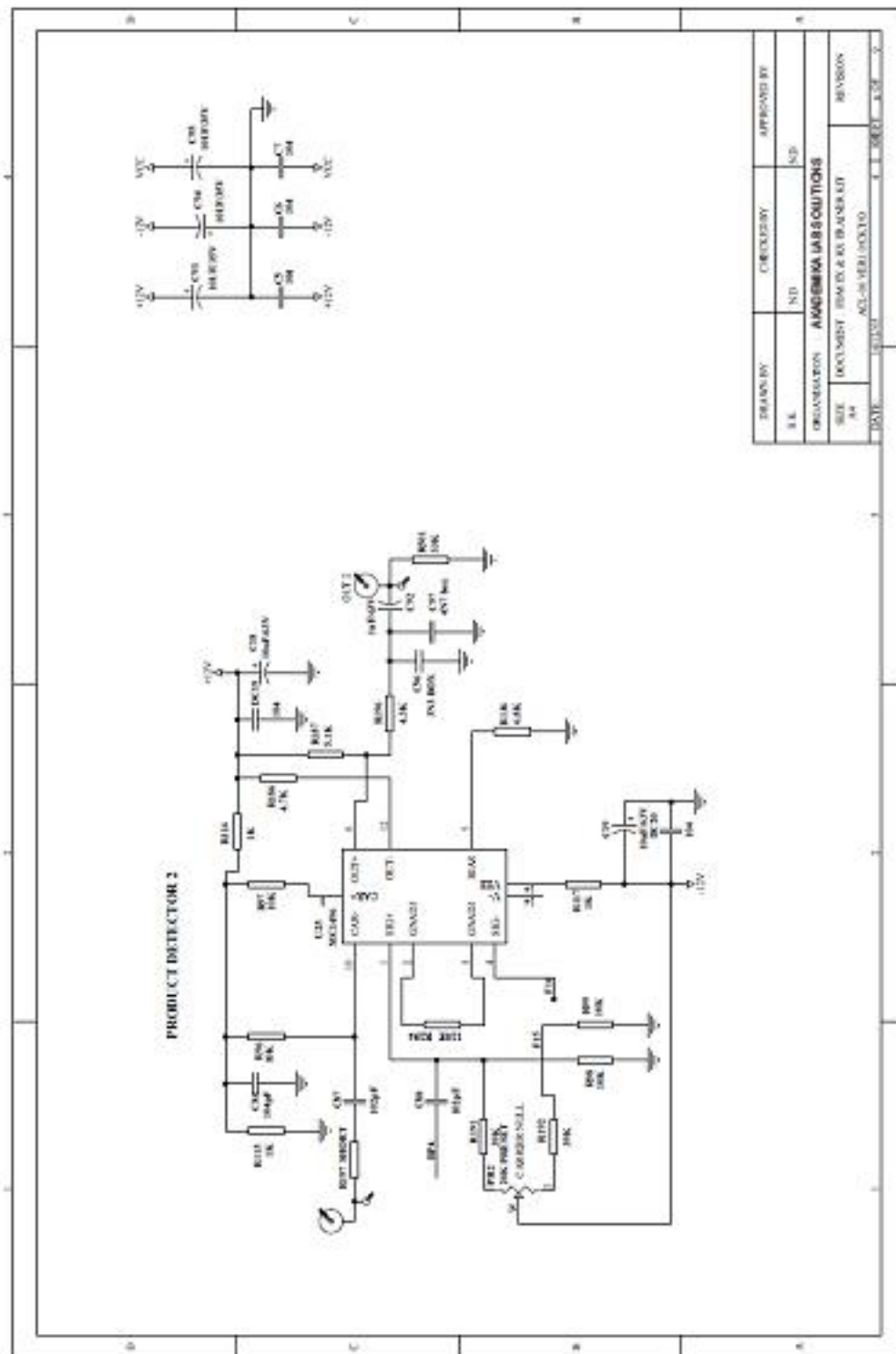
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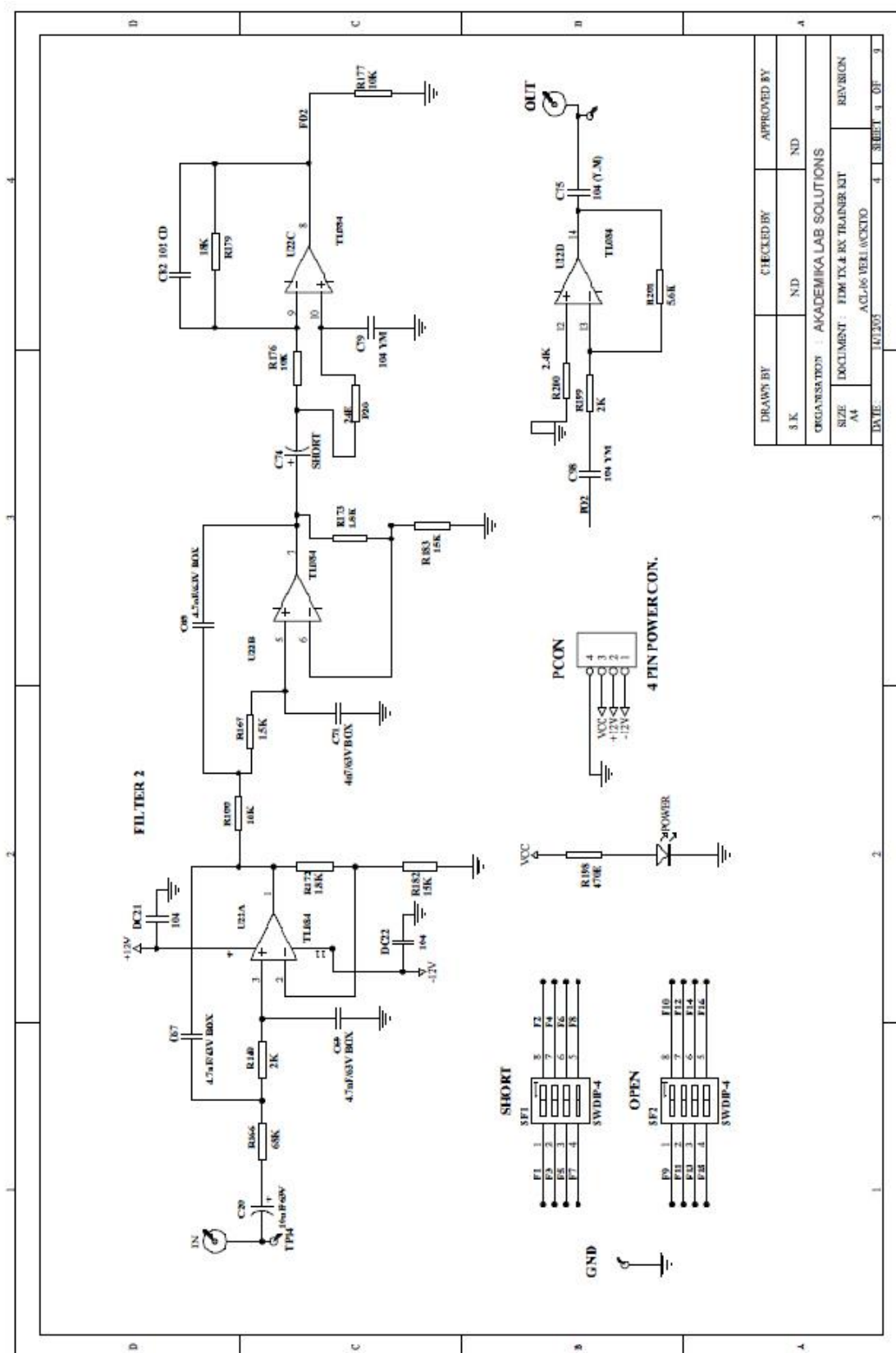












EXPERIMENT NO.1



SWITCH FAULTS

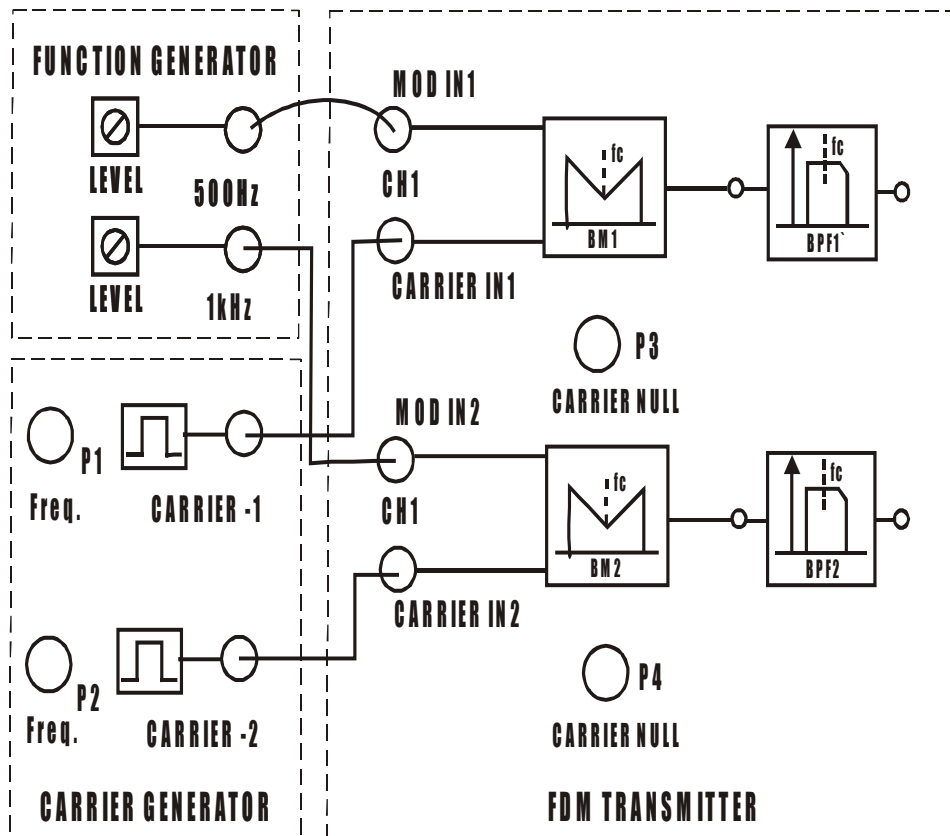


FIG 1.1 BLOCK DIAGRAM FOR SSB MODULATION

EXPERIMENT NO: 1

NAME:-

To Study of Amplitude Modulation With and Without Suppressed Carrier

OBJECTIVE:-

The objective of this experiment is to study the operation of an AM modulation with and without suppressed carrier.

THEORY:-

The amplitude-modulation technique called double-sideband/ suppressed-carrier (DSB-SC) has an important advantage over regular AM (DSB-FC). The carrier is suppressed during the modulation process. As a result, most of the power in a regular AM transmission, which provides no information, is eliminated.

As will be obvious by simple inspection of the waveform of a DSB-SC signal, an AM rectifier cannot be used to demodulate DSB-SC. Demodulation can be achieved only if a locally generated carrier signal is introduced. It must not only have exactly the correct frequency (be frequency-coherent) but also have a phase very close to what the carrier would have if it had been transmitted; that is, DSB-SC demodulation must also be approximately phase-coherent.

The peak detector discussed for demodulation of regular AM (DSB-FC) will not yield the correct result for DSB-SC. For instance, when the input is the sinusoidal tone-modulated DSB-SC signal, the output of a peak detector will be the "cusp" signal of Figure a.

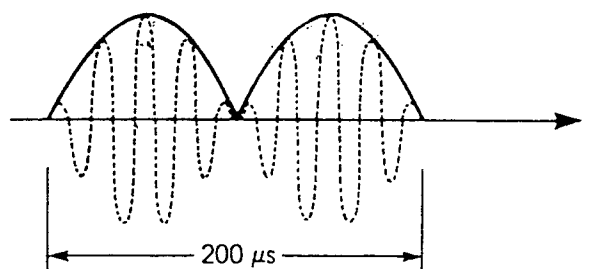


Figure a. Result of non-coherent demodulation of DSB-SC

EQUIPMENTS:-

- ACL-FDM Kit
- Dual Channel Function Generator
- Patch Cords
- Power Supply (Use only one provided)
- Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE:-

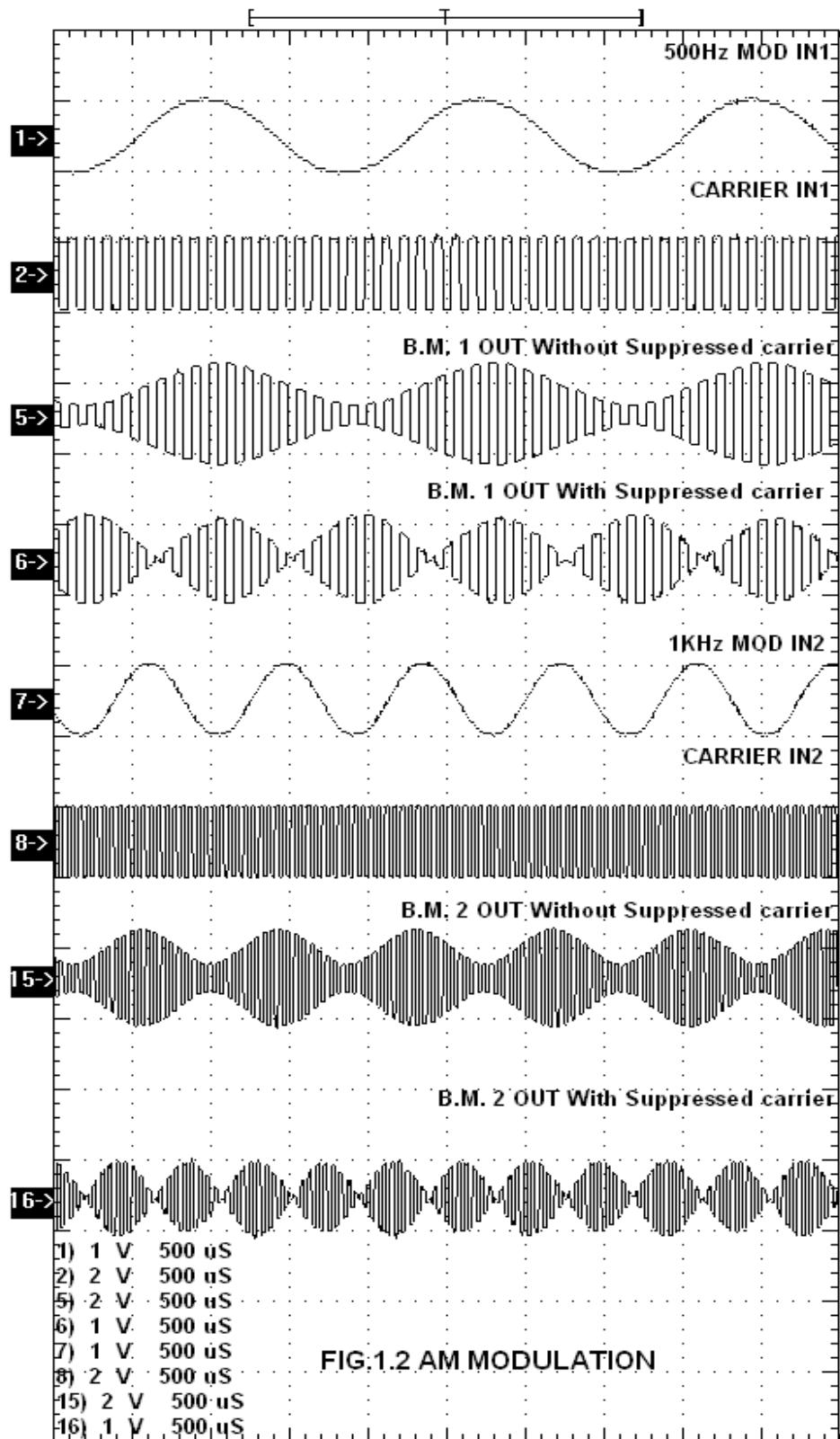
- Refer to the Block diagram and carry out the following connections and switch settings.

- Connect the power supply cables with proper polarity to ACL-FDM kit. While connecting this, ensure that the power supply is OFF.
- Switch on the power supply.
- Connect the **500Hz, 1Vpp** signal from external function generator as a constant signal to the **MOD IN 1** post of the FDM Transmitter of ACL-FDM.
- Connect the onboard carrier frequency **9KHz, 2Vpp** to the **CARRIER IN 1** post of the FDM Transmitter.
- Keep the **CARRIER NULL** at center position by rotating the pot **P3**, so that the modulator1 is “balanced” and an AM signal with suppressed carrier, as shown in waveform, is obtained across the output whose amplitude is about 1Vpp.
- Connect the output of Balanced Modulator1 to the input of Band Pass Filter1.
- Observe the DSB-SC output at the given output test point of the Band Pass Filter1.
- Repeat the same procedure for Balanced Modulator2.
- Connect the **1KHz, 1Vpp** signal from external function generator as a constant signal to the **MOD IN 2** post of the FDM Transmitter of ACL-FDM.
- Connect the onboard carrier frequency **20KHz, 2Vpp** to the **CARRIER IN 2** post of the FDM Transmitter.
- Keep the **CARRIER NULL** at center position by rotating the pot **P4**, so that the modulator2 is “balanced” and an AM signal with suppressed carrier, as shown in waveform, is obtained across the output whose amplitude is about 1Vpp.
- Connect the output of Balanced Modulator2 to the input of Band Pass Filter2.
- Observe the DSB-SC output at the given output test point of the Band Pass Filter2.

OBSERVATION:-

Observe the following waveforms on oscilloscope and plot on the paper.

- Modulating signal 1 500Hz at **MOD IN 1**
- Carrier signal 1 at **CARRIER IN 1**
- Balanced Modulator 1 output without suppressed carrier at **BM1** output test point
- Balanced Modulator 1 output with suppressed carrier at **BM1** output test point
- Modulating signal 2 1KHz at **MOD IN 2**
- Carrier signal 2 at **CARRIER IN 2**
- Balanced Modulator 2 output without suppressed carrier at **BM2** output test point
- Balanced Modulator 2 output with suppressed carrier at **BM2** output test point



EXPERIMENT NO.2

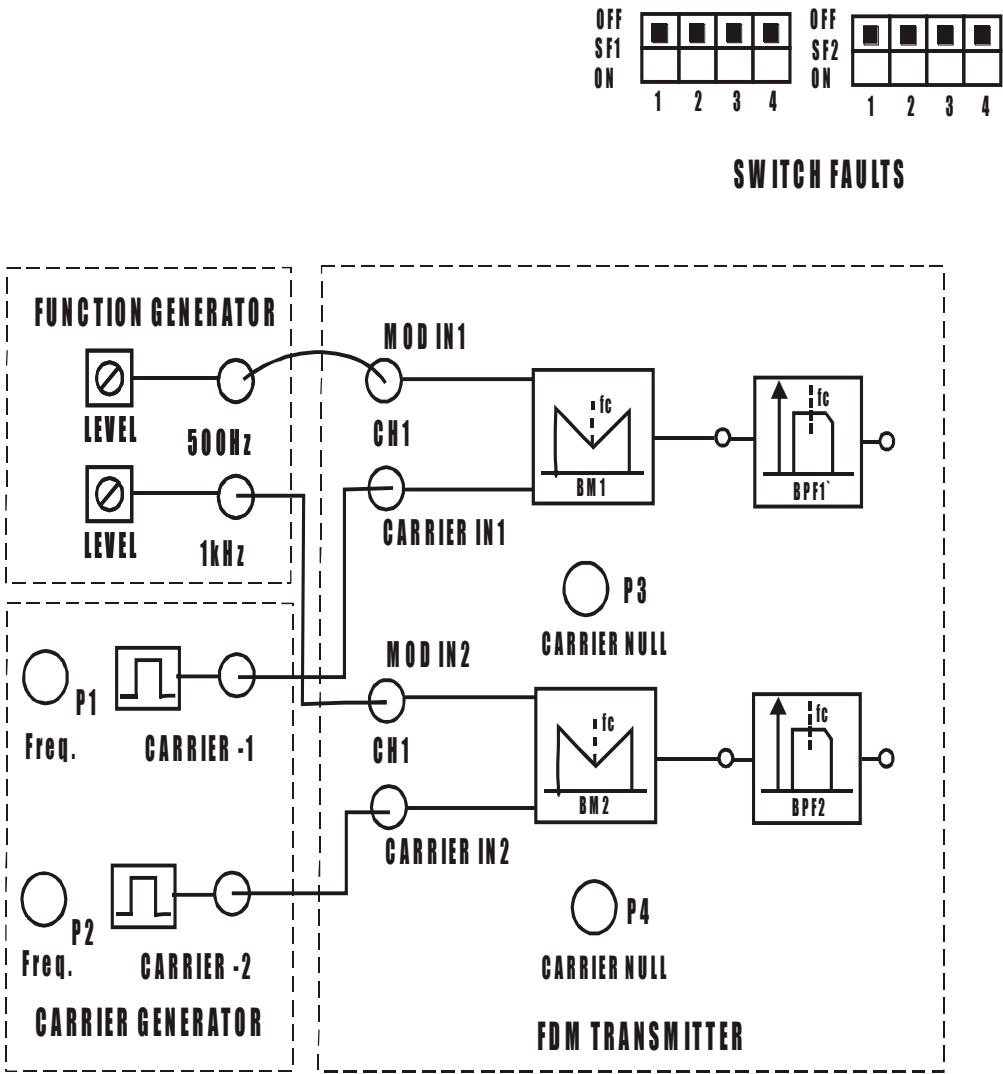


FIG 2.1 BLOCK DIAGRAM FOR FREQUENCY RESPONSE OF BANDPASS FILTERS

EXPERIMENT NO. 2

NAME:-

To Study of Frequency Response of Band Pass Filters

OBJECTIVE:-

The objective of this experiment is to study frequency response of band pass filters.

THEORY:-

A band pass filter has a pass band between two cut off frequencies f_H and f_L such that $f_H > f_L$. Any input frequency outside this pass-band is attenuated. Basically there are two types of band pass filters:

- (1) Wide band pass
- (2) Narrow band pass

A filter is said to be wide band pass if its figure of merit or quality factor $Q < 10$. If $Q > 10$ then it will be narrow band pass filter. Thus Q is a measure of selectivity, meaning that higher the value of Q , more selective is the filter or narrower is its bandwidth (BW). The relationship between Q , 3-dB bandwidth and the center frequency f_c is given by

$$Q = f_c / BW = f_c / (f_H - f_L)$$

For the wide band pass filter the center frequency f_c can be defined as

$$f_c = (f_H f_L)^{1/2}$$

where f_H = high cut off frequency (Hz)

f_L = low cut off frequency of the wide band pass filter (Hz)

EQUIPMENTS:-

- ACL-FDM Kit
- Dual Channel Function Generator
- Patch Chords
- Power Supply (Use only one provided)
- Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

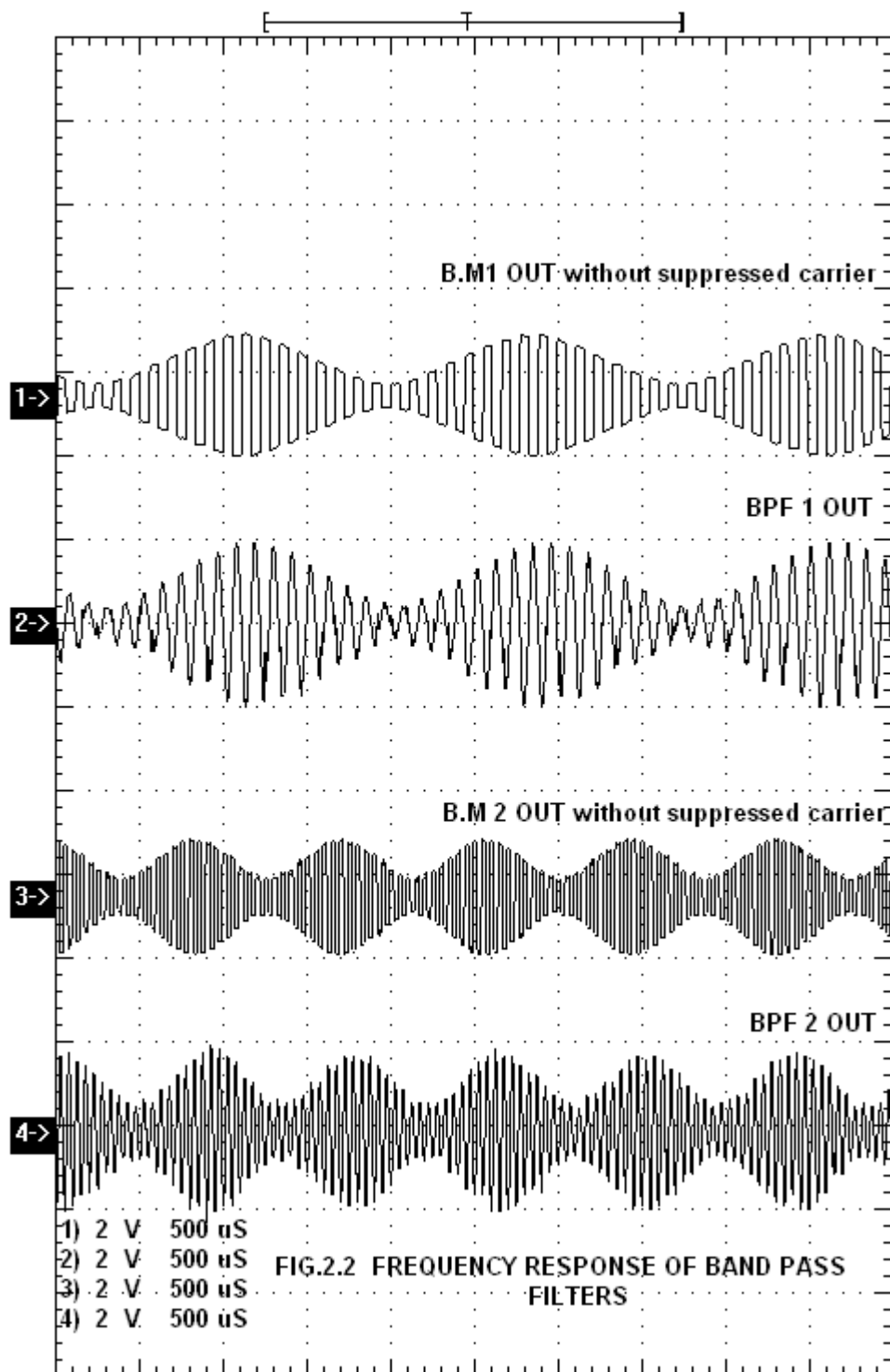
PROCEDURE:-

- Refer to the Block diagram and carry out the following connections and switch settings.
- Connect the power supply cables with proper polarity to ACL-FDM kit. While connecting this, ensure that the power supply is OFF.
- Switch on the power supply.
- Connect the **500Hz, 1Vpp** signal from external function generator as a constant signal to the **MOD IN 1** post of the FDM Transmitter of ACL-FDM.
- Connect the onboard carrier frequency **9KHz, 2Vpp** to the **CARRIER IN 1** post of the FDM Transmitter.
- Keep the **CARRIER NULL** fully clockwise or anticlockwise by rotating the pot **P3**, so that the **BM1** is “unbalanced” and an AM signal with not suppressed carrier, as shown in waveform, is obtained across the output whose amplitude is about 1Vpp.
- Observe the AM output at the given output test point of the Band Pass Filter1, as shown in waveform.
- Vary the frequency of CARRIER-1 from 5KHz to 15KHz using pot **P1**. List the deviations of the output amplitude versus carrier frequency in the table.
- Note the frequency where you get the maximum output. This is the Center Frequency (f_c) of Band Pass Filter1.
- Repeat the same procedure for Band Pass Filter2.
- Connect the **1KHz, 1Vpp** signal from external function generator to the **MOD IN 2** post of the FDM Transmitter of ACL-FDM.
- Connect the onboard carrier frequency **20KHz, 2Vpp** to the **CARRIER IN 2** post of the FDM Transmitter.
- Keep the **CARRIER NULL** fully clockwise or anticlockwise by rotating the pot **P4**, so that the **BM2** is “unbalanced” and an AM signal with not suppressed carrier, as shown in waveform, is obtained across the output whose amplitude is about 1Vpp.
- Observe the AM output at the given output test point of the Band Pass Filter2, as shown in waveform.
- Vary the frequency of CARRIER-2 from 15KHz to 25KHz using pot **P2**. List the deviations of the output amplitude versus carrier frequency in the table.
- Note the frequency where you get the maximum output. This is the Center Frequency (f_c) of Band Pass Filter2.

OBSERVATION:-

Observe the following waveforms on oscilloscope and plot on the paper.

- Balanced Modulator 1 output without suppressed carrier at **BM1** output test point
- Band Pass Filter 1 output at **BPF1** output test point
- Balanced Modulator 2 output without suppressed carrier at **BM2** output test point
- Band Pass Filter 2 output at **BPF2** output test point



EXPERIMENT NO.3

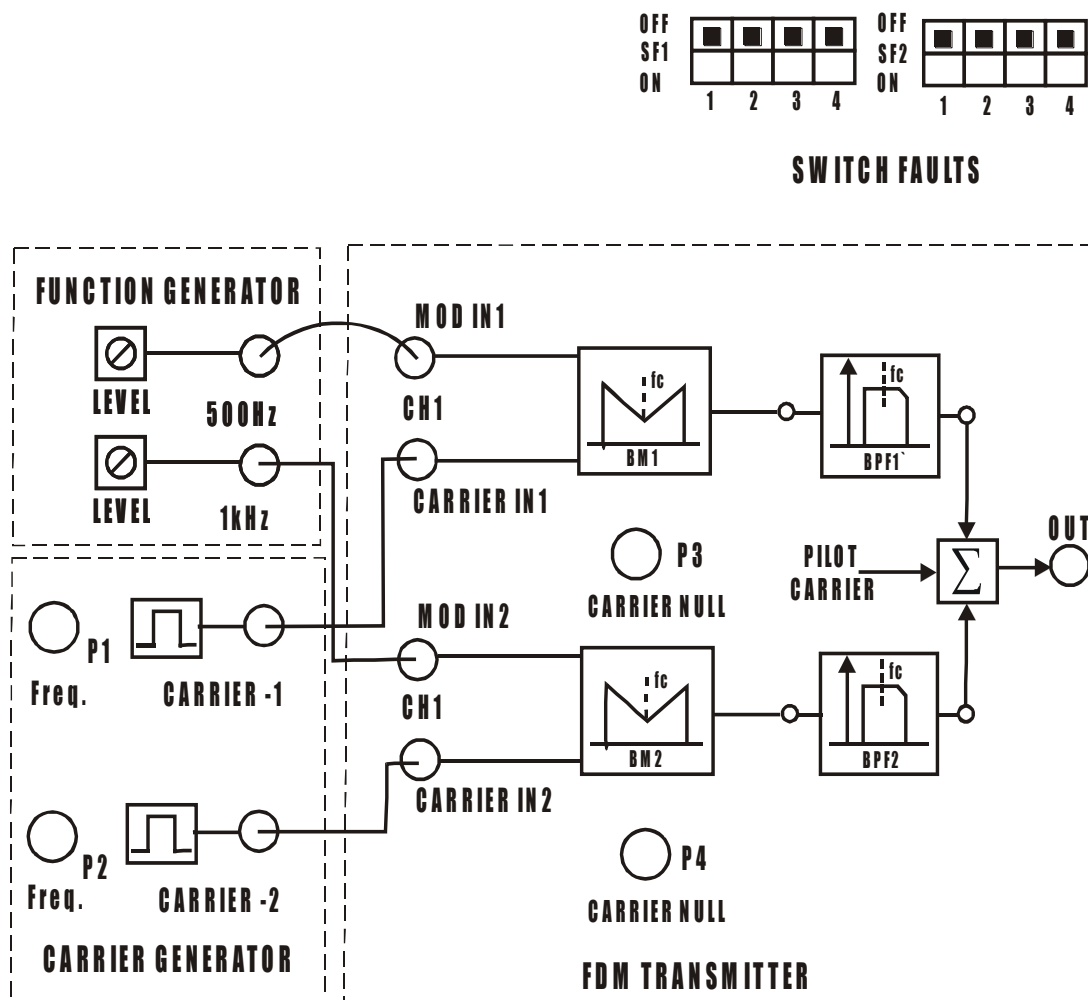


FIG.3.1 BLOCK DIAGRAM FOR FREQUENCY DIVISION MULTIPLEXING

EXPERIMENT NO. 3

NAME:-

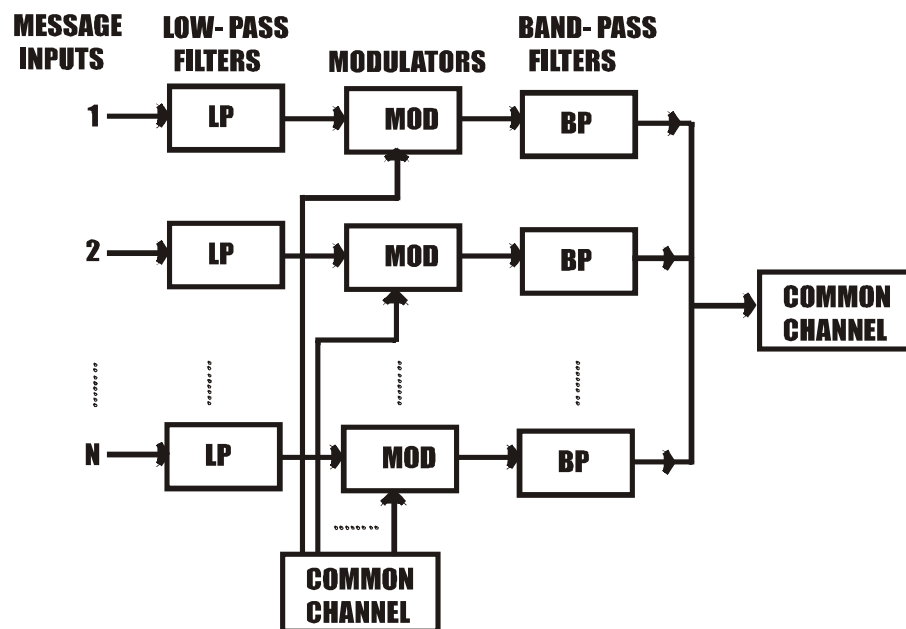
To Study Operation of Frequency Division Multiplexing

OBJECTIVE:-

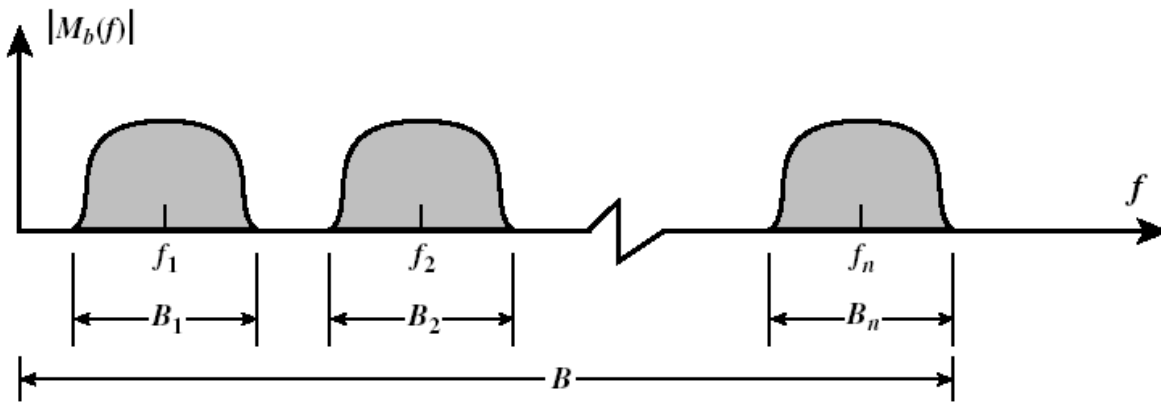
The objective of this experiment is to study the Frequency Division Multiplexing.

THEORY:-

Multiplexing is a signal processing operation by which a number of independent messages can be combined into a composite signal suitable for transmission over common channel. The technique of separating these messages in frequency is referred to as frequency division multiplexing (FDM).



(a) FDM TRANSMITTER



(b) Spectrum of composite baseband modulating signal

A block diagram of an FDM transmitter is shown in FIG. (a). The incoming message signals are assumed to be of low-pass type, but their spectra do not necessarily have nonzero values all the way down to zero frequency. Following each signal input, we have shown a low pass filter, which is designed to remove high frequency components that do not contribute significantly to signal representation but are capable of disturbing other message signals that share the common channel. The low pass filters may be omitted only if the input signals are sufficiently band limited initially. The filtered signal is then applied to modulators. Modulation technique used here is amplitude modulation technique, in which the amplitude of high frequency sine wave (carrier) is varied in accordance with the instantaneous value of the modulating signal. The modulators thus shift the frequency ranges of the signals so as to occupy mutually exclusive frequency intervals. The necessary carrier frequencies needed to perform these frequency translations are obtained from a carrier supply. The band-pass filters following the modulators are used with respect to the band of each modulated wave to its prescribed range. The resulting band pass filter outputs are combined in parallel to form the input to the common channel. Spectrum of composite base-band modulating signal is shown in FIG. (b).

EQUIPMENTS:-

- ACL-FDM Kit
- Dual Channel Function Generator
- Patch Chords
- Power Supply (Use only one provided)
- Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

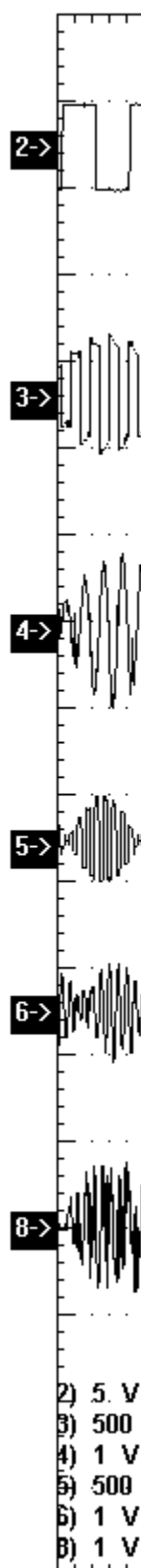
PROCEDURE:-

- Refer to the block diagram and carry out the following connections and switch settings.
- Connect the power supply cables with proper polarity to ACL-FDM kit. While connecting this, ensure that the power supply is OFF.
- Switch on the power supply.
- Connect the **500Hz,1Vpp** Signal from external function generator to the **MOD IN 1** post of ACL-FDM.
- Connect the **1KHz, 1Vpp** Signal from external function generator to the **MOD IN 2** post of ACL-FDM.
- Set carrier frequency **CARRIER 1** to **9KHz** using pot **P1**, set carrier frequency **CARRIER 2** to **20KHz** using pot **P2**.
- Connect the **CARRIER 1** post of the Carrier Generator to the **CARRIER IN1** post of FDM Transmitter and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Transmitter.
- Keep the **CARRIER NULL** at center position by rotating the pot **P3** and **P4**, so that both the modulators are “balanced” and an AM signal with suppressed carrier, as shown in waveform, is obtained across the output whose amplitude is about 1Vpp.
- Observe the signal at the given out put test point of BM1 and BM2 as shown in fig.
- Observe the Band Pass Filter-1 output at the given output test point, as shown in waveform. There should be maximum signal transmission at center frequency (f_c) of about 9KHz.
- Observe the Band Pass Filter-2 output at the given output test point, as shown in waveform. There should be maximum signal transmission at center frequency (f_c) of about 20KHz.
- Observe the FDM signal at the **OUT** post of the FDM Transmitter block. Observe the waveform, which is frequency division multiplexed at frequency **256KHz** of pilot carrier.

OBSERVATION:-

Observe the following waveforms on oscilloscope and plot on the paper.

- Pilot Carrier at **CLK**
- Balanced Modulator 1 output with suppressed carrier at **BM1** output test point
- Band Pass Filter 1 output at **BPF1** output test point
- Balanced Modulator 2 output with suppressed carrier at **BM2** output test point
- Band Pass Filter 2 output at **BPF2** output test point
- Frequency Division Multiplexed output at **FDM OUT**



EXPERIMENT NO.4

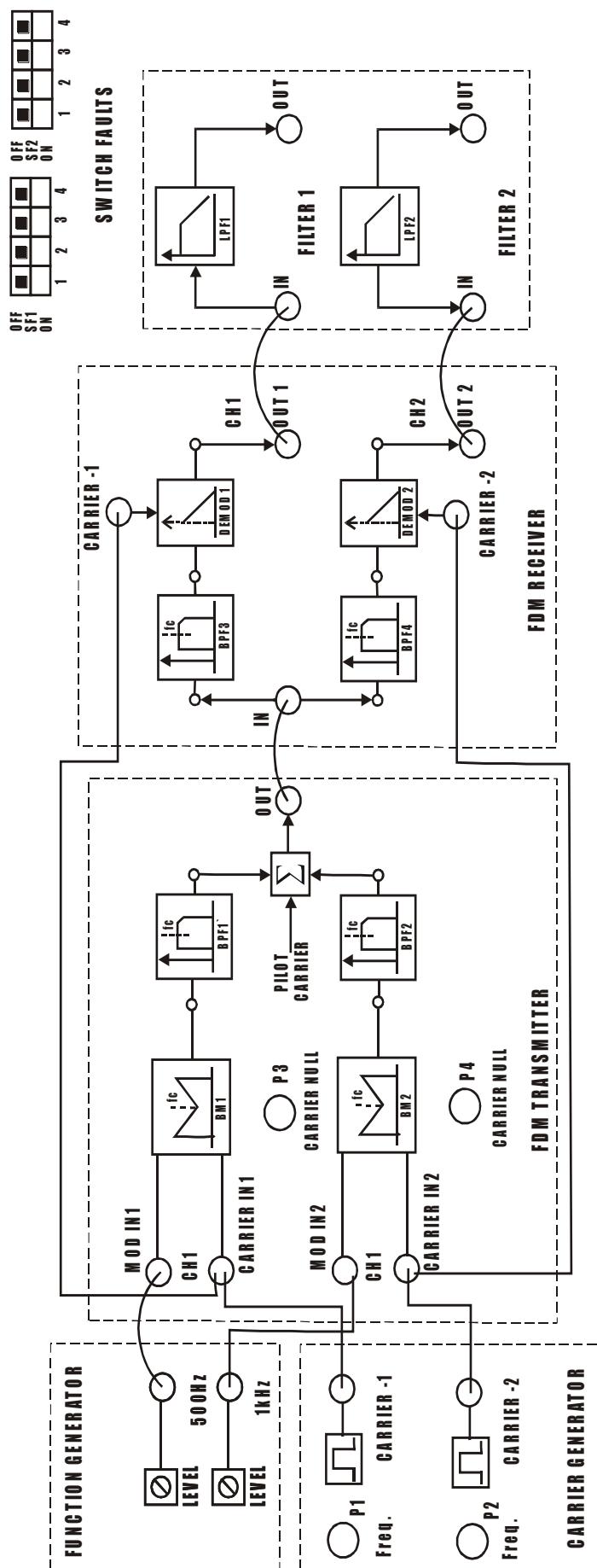


FIG. 4.1 BLOCK DIAGRAM FOR FREQUENCY DIVISION DEMULTIPLEXING

EXPERIMENT NO. 4

NAME:-

To Study Operation of Frequency Division De-multiplexing

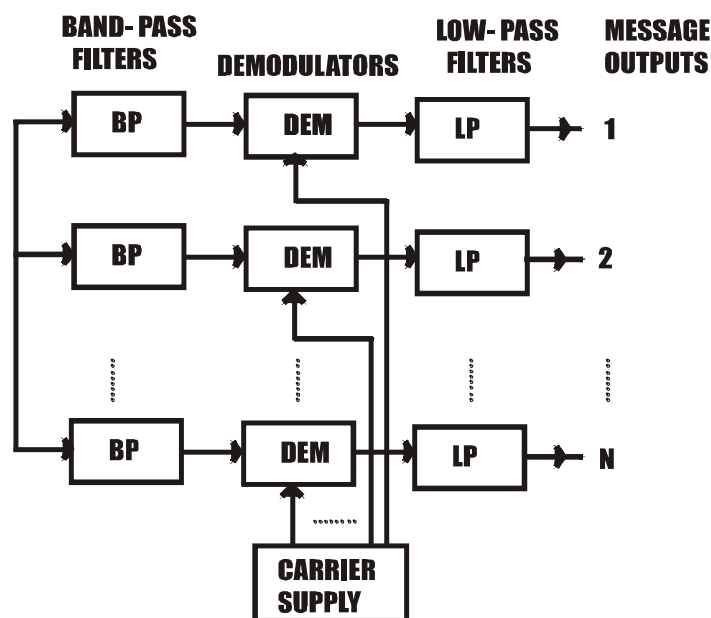
OBJECTIVE:-

The objective of this experiment is to study the Frequency Division De-multiplexing.

THEORY:-

Message recovery or demodulation of FDM is accomplished by using a product detector. The product detector generates the output with the product of the two input signals. The carrier signal and the amplitude-modulated signal are multiplied. The multiplication of two signals produces different components. The signal, which corresponds to the detected signal, is separated via the low pass filter.

Note that FDM shown in FIG. (c) Operates in only one direction. To achieve two way transmission, as in telephony, for e.g. we have to completely duplicate the de-multiplexing facilities, with components connected in reverse order & with the signal waves preceding from right to left.



(c) FDM RECEIVER

At FDM receiver, as shown in FIG (c), a bank of band-pass filters, with their inputs connected in parallel, is used to separate the message signals on a frequency-occupancy basis. Finally, the original message signals are recovered by individual demodulators.

EQUIPMENTS:-

- ACL-FDM Kit
- Dual Channel Function Generator
- Patch Cords
- Power Supply (Use only one provided)
- Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE:-

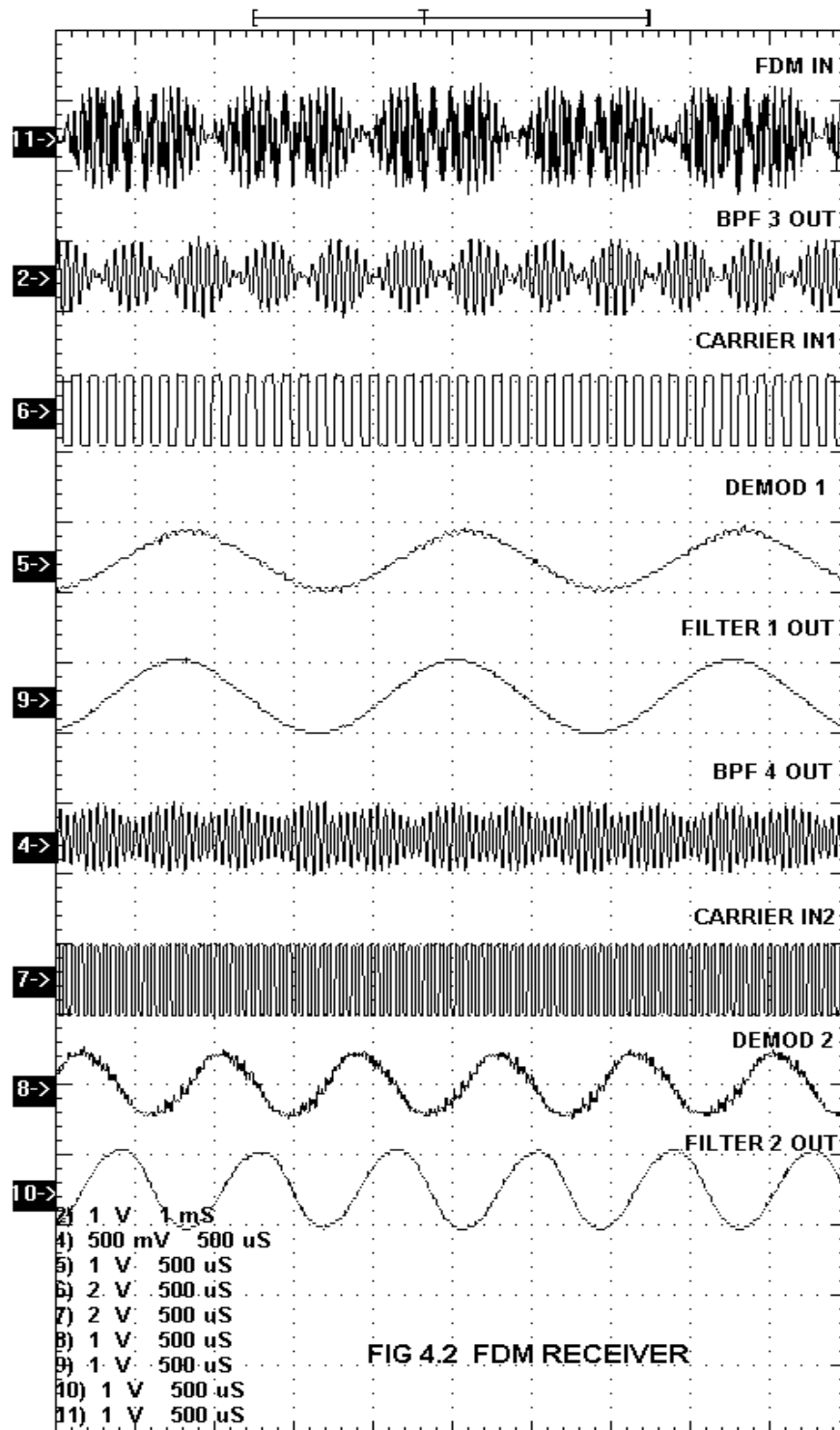
- Refer to the Block diagram and carry out the following connections and switch settings.
- Connect the power supply cables with proper polarity to ACL-FDM kit. While connecting this, ensure that the power supply is OFF.
- Switch on the power supply.
- For de-multiplexing, keep the connections same as given in the procedure steps 5 to 13 of experiment no. 3 and then continue with the following procedure.
- Connect the FDM signal from the **OUT** post of the FDM Transmitter to the **IN** post of the FDM Receiver.
- Connect the **CARRIER 1** post of the CARRIER GENERATOR to the **CARRIER IN1** post of FDM Receiver and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Receiver.
- Observe the Band pass filter response at the given output test points of **BPF3** and **BPF4** of the FDM Receiver.
- Observe the demodulated output at the posts **OUT1** and **OUT2** of the FDM Receiver.
- Connect the demodulated output from the **OUT1** post of the FDM Receiver to the **IN** post of the Filter 1 and the **OUT2** post of the FDM Receiver to the **IN** post of the Filter 2.
- Observe the recovered filtered sine wave at both the **OUT** posts of Filter 1 and Filter 2.

OBSERVATION:-

Observe the following waveforms on oscilloscope and plot on the paper.

- Frequency Division Multiplexed signal at **IN** post of FDM Receiver
- Band Pass Filter 3 output at **BPF3** output test point
- Carrier signal 1 at **CARRIER IN 1**
- Demodulated signal at **OUT 1**
- Recovered signal at **OUT** of FILTER 1

- Band Pass Filter 4 output at **BPF4** output test point
- Carrier signal 2 at **CARRIER IN 2**
- Demodulated signal at **OUT 2**
- Recovered signal at **OUT** of FILTER 2



EXPERIMENT NO.5

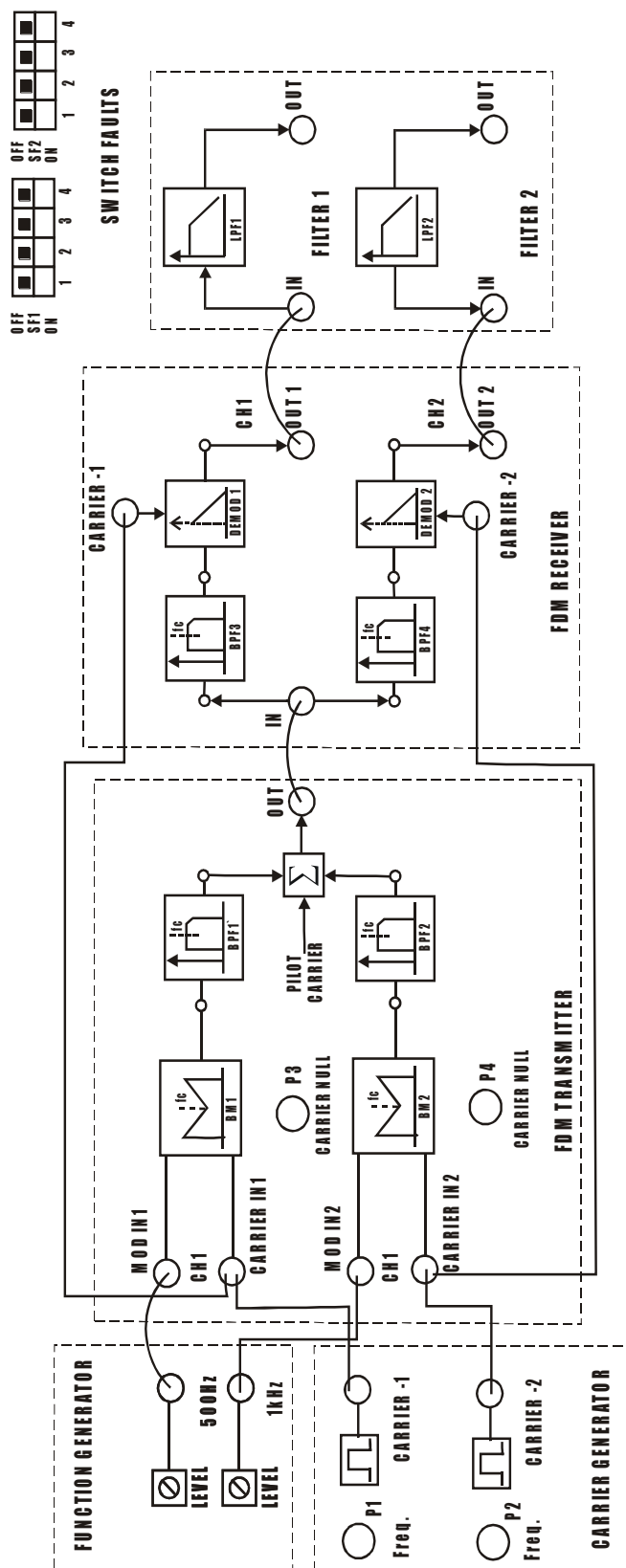


FIG. 5.1 BLOCK DIAGRAM FOR OUT OF BAND SIGNALLING

EXPERIMENT NO.5

NAME:-

To Study Effect Of Out Of Band Signaling

OBJECTIVE:-

The objective of this experiment is to study the effect of out of band signaling.

THEORY:-

In telecommunications, the term out-of-band signaling has the following meanings.

- Signaling that uses a portion of the channel bandwidth provided by the transmission medium, e.g. the carrier channel which portion is above the highest frequency used and is denied to the speech or intelligence path by filters.
Out-of-band signaling results in a lowered f_H of the effective available bandwidth.
- Signaling via a different channel (either FDM or TDM) from that used for the primary information transfer. Example: SS7.
Theoretically the output of the out-of-band detector is at high level when signaling occurs and at low level when no signaling is recorded.

EQUIPMENTS:-

- ACL-FDM Kit
- Dual Channel Function Generator
- Patch Chords
- Power Supply (Use only one provided)
- Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE:-

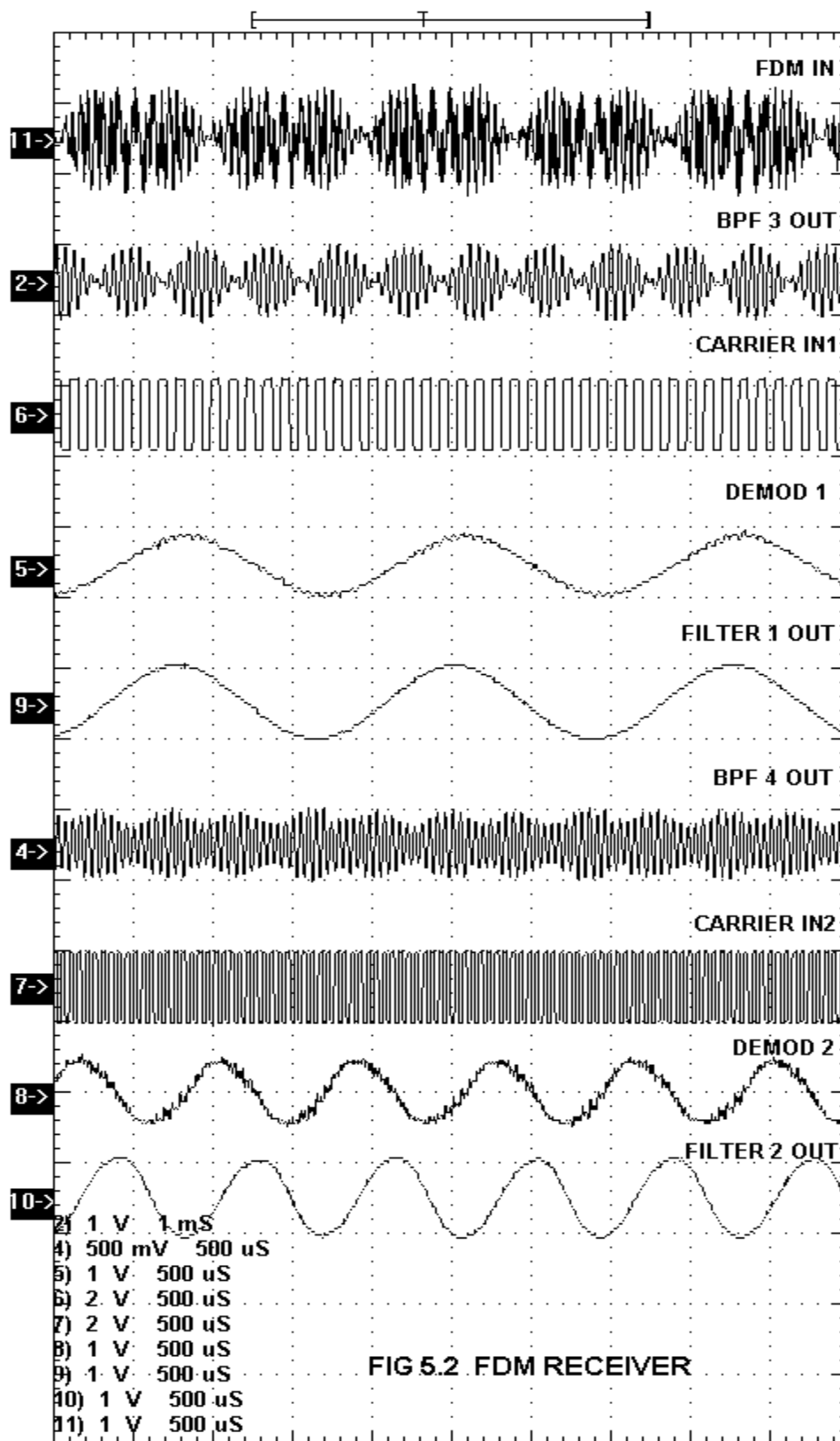
- Refer to the Block diagram and carry out the following connections and switch settings.
- Connect the power supply cables with proper polarity to ACL-FDM kit. While connecting this, ensure that the power supply is OFF.
- Switch on the power supply.
- Connect the **500Hz, 1Vpp** Signal from external function generator to the **MOD IN 1** post of ACL-FDM.
- Connect the **1KHz, 1Vpp** Signal from external function generator to the **MOD IN 2** post of ACL-FDM.

- Set carrier frequency **CARRIER 1** to **9KHz** using pot **P1**, set carrier frequency **CARRIER 2** to **20KHz** using pot **P2**.
- Connect the patch cord from **CARRIER 1** post of the Carrier Generator to the **CARRIER IN1** post of FDM Transmitter and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Transmitter.
- Keep the **CARRIER NULL** at center position by rotating the pot **P3** and **P4**, so that both the modulators are “balanced” and an AM signal with suppressed carrier, as shown in waveform, is obtained across the output whose amplitude is about 1Vpp.
- Observe the signal at the given out put test point of **BM1** and **BM2** as shown in fig.
- Observe the Band Pass Filter-1 output at the given output test point, as shown in waveform. There should be maximum signal transmission at center frequency (Fc) of about 9KHz, 2Vpp.
- Observe the Band Pass Filter-2 output at the given output test point, as shown in waveform. There should be maximum signal transmission at center frequency (Fc) of about 20KHz, 2Vpp.
- Observe the FDM signal at the **OUT** post of the FDM Transmitter block. Observe the waveform, which is frequency division multiplexed at frequency **256KHz** of pilot carrier.
- Connect the FDM signal from the **OUT** post of the FDM Transmitter to the **IN** post of the FDM Receiver.
- Connect the patch cord from **CARRIER 1** post of the CARRIER GENERATOR to the **CARRIER IN1** post of FDM Receiver and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Receiver.
- Observe the Band pass filter response at the given output test points of **BF3** and **BF4** of the FDM Receiver.
- Observe the demodulated output at the out posts **OUT1** and **OUT2** of the FDM Receiver.
- Connect the demodulated output from the **OUT1** post of the FDM Receiver to the **IN** post of the Filter 1 and the **OUT2** post of the FDM Receiver to the **IN** post of the Filter 2.
- Observe the recovered filtered sine wave at both the **OUT** posts of Filter 1 and Filter 2.
- Now vary the carrier frequency by varying pot **P1** of the Carrier Generator and observe the effect of out of band signaling on the recovered signal at the **OUT** post of Filter 1.
- Similarly, vary the carrier frequency by varying pot **P2** of the Carrier Generator and observe the effect of out of band signaling on the recovered signal at the **OUT** post of Filter 2.

OBSERVATION:-

Observe the following waveforms on oscilloscope and plot on the paper.

- Frequency Division Multiplexed signal at **IN** post of FDM Receiver
- Band Pass Filter 3 output at **BPF3** output test point
- Carrier signal 1 at **CARRIER IN 1**
- Demodulated signal at **OUT 1**
- Recovered signal at **OUT** of FILTER 1
- Band Pass Filter 4 output at **BPF4** output test point
- Carrier signal 2 at **CARRIER IN 2**
- Demodulated signal at **OUT 2**
- Recovered signal at **OUT** of FILTER 2



EXPERIMENT NO.6

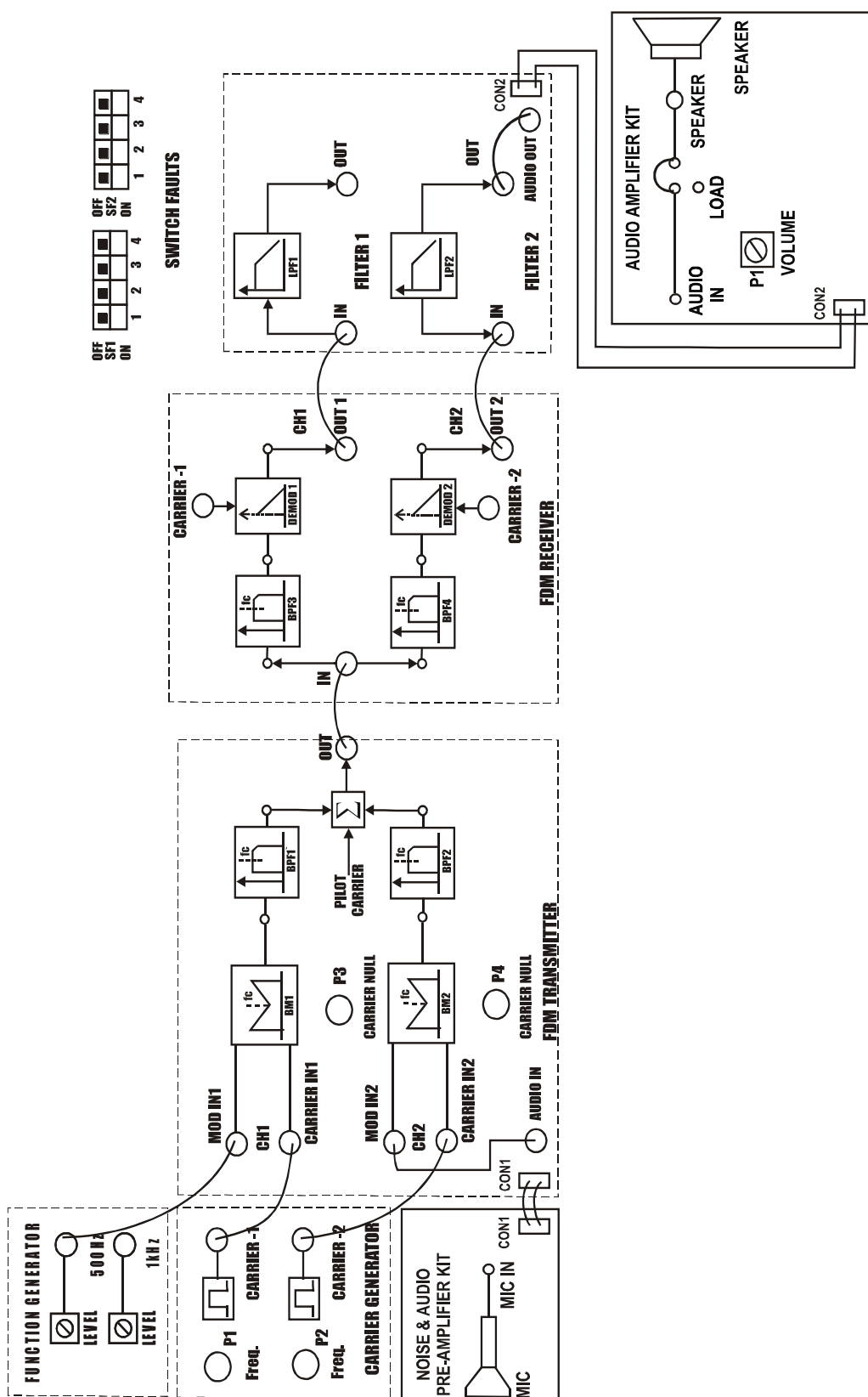


FIG. 6.1 BLOCK DIAGRAM FOR THE VOICE COMMUNICATION THROUGH FDM

EXPERIMENT NO.6

NAME:-

To Study Voice Communication through FDM (Optional)

OBJECTIVE:-

The objective of this experiment is to study the voice communication through FDM.

THEORY:-

Voice being analog signal is sampled by using any one method at transmitter end and reconstructed using filter of different order like the 2nd and 4th order low pass butter-worth filter at the receiver side. For performing voice communication, set up the basic Frequency Division Multiplexing and De-multiplexing experiment as mentioned earlier and then feed voice signal as input and the reconstructed output to the speaker.

EQUIPMENTS:-

- ACL-FDM Kit
- Dual Channel Function Generator
- Patch Chords
- Audio Input Output Kit
- Power Supply (Use only one provided)
- Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE:-

- Refer to the Block diagram and carry out the following connections and switch settings.
- Connect the **CON1** post of Noise and Audio Pre-Amplifier kit to the **CON1** post of ACL-FDM kit using power connection cable.
- Connect the **CON2** post of Audio Amplifier kit to the **CON2** post on ACL-FDM kit using power connection cable.
- Connect the power supply in proper polarity to the ACL-FDM kit.
- Switch on the power supply.
- Connect the Microphone to the **MIC IN** post of Noise and Audio Pre-Amplifier kit.
- Connect **500Hz, 1Vpp** Signal from external function generator to the **MOD IN 1** post of ACL-FDM.
- Connect the audio signal from the **AUDIO IN** post to the **MOD IN 2** post of ACL-FDM.

- Set carrier frequency **CARRIER 1** to **9KHz** using pot **P1**. Set carrier frequency **CARRIER 2** to **20KHz** using pot **P2**.
- Connect the **CARRIER 1** post of the Carrier Generator to the **CARRIER IN1** post of FDM Transmitter and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Transmitter.
- Keep the **CARRIER NULL** at center position by rotating the pot **P3** and **P4**, so that both the modulators are “balanced” and an AM signal with suppressed carrier is obtained across the output whose amplitude is about 1Vpp.
- Connect the FDM signal from the **OUT** post of the FDM Transmitter to the **IN** post of the FDM Receiver.
- Connect the **CARRIER 1** post of the CARRIER GENERATOR to the **CARRIER IN1** post of FDM Receiver and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Receiver.
- Connect the demodulated output from the **OUT1** post of the FDM Receiver to the **IN** post of the Filter 1 and the **OUT2** post of the FDM Receiver to the **IN** post of the Filter 2.
- Put the jumper in Audio Amplifier kit towards speaker selection as shown in fig.
- Connect the speaker provided with the kit to the speaker jack on Audio Amplifier kit.
- Then connect the output signal from **OUT** post of the Filter 2 to the **AUDIO OUT** post and check the voice link.

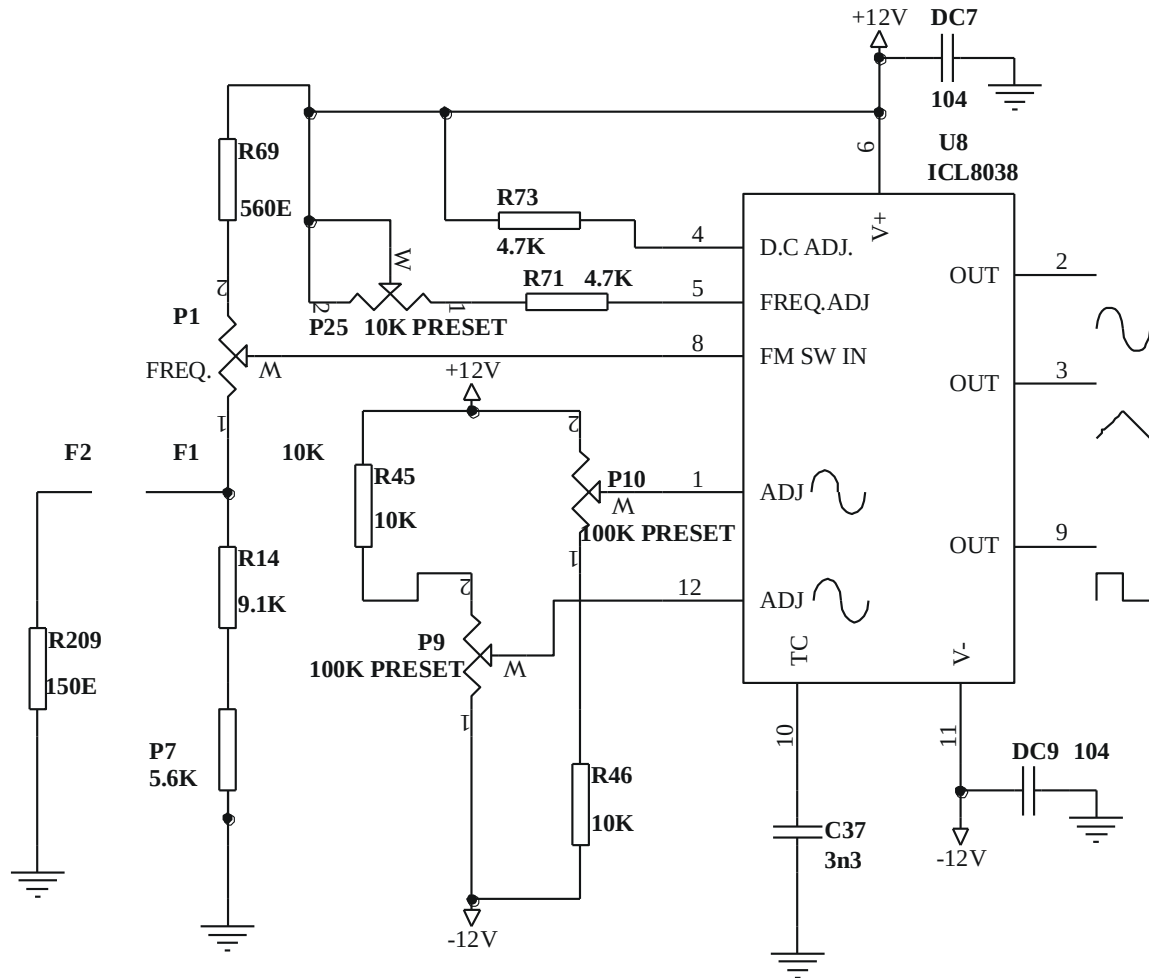
SWITCHFAULTS:-

Switch fault section on ACL-FDM is provided with 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit. In order to keep ACL-FDM kit fully operational, all switches should be set to OFF position before use. Switch faults are introduced one at a time.

1. Effect Of Switch Fault No. 1 In Carrier Generator 1 Section

DESCRIPTION:-

The ICL8038 waveform generator is a monolithic integrated circuit capable of producing high accuracy sine, square, triangular saw tooth and pulse waveforms with a minimum of external components. The sine wave is created by feeding the triangle wave into a nonlinear network (sine converter). This circuit generates a carrier frequency varying from 1 KHz to 20 KHz with amplitude 2Vpp.

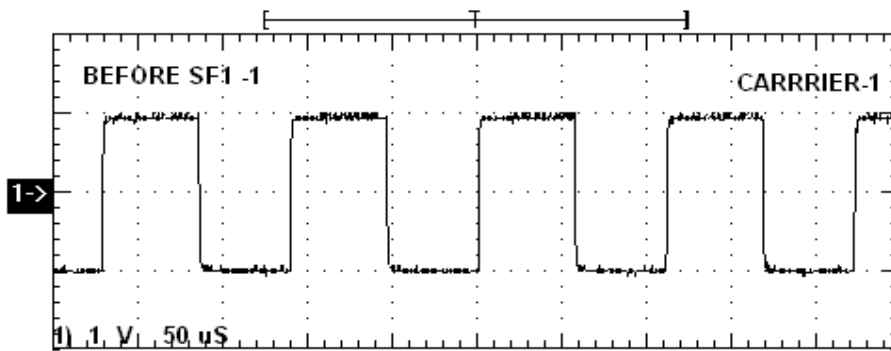


PROCEDURE:-

First keep switch **1** of **SF1** to OFF position.

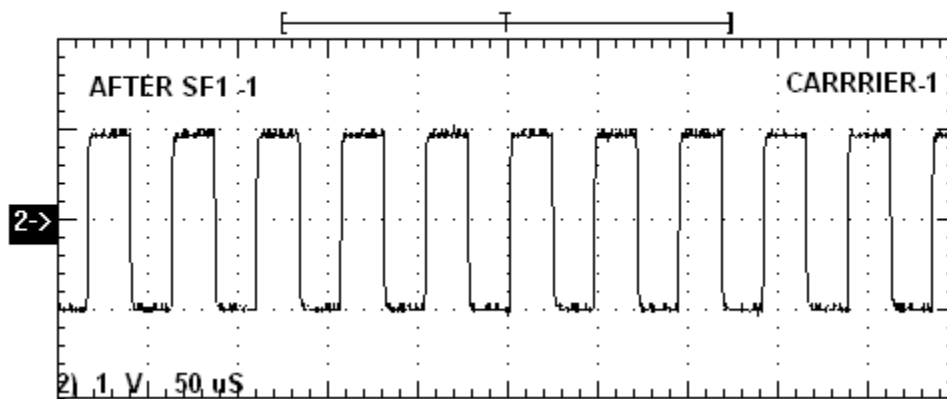
- Connect power supply as per polarity and switch it on.
- Observe the waveform at the given **CARRIER-1** post of the Carrier Generator.

Waveform before Switch Fault



Now put the switch **1** of **SF1** in the Switch Fault to ON position and observe its effect.

Waveform after Switch Fault

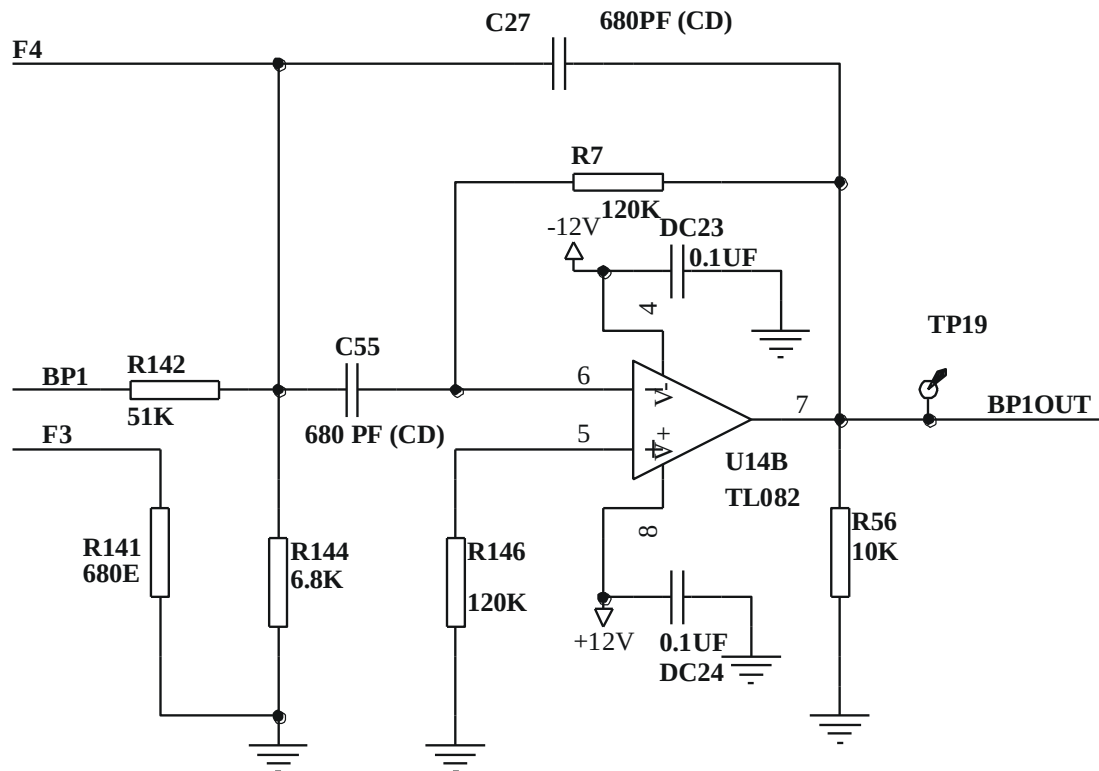


Action and Effect

When we put the switch **1** of **SF1** in Switch Fault section to ON position, resistor R209 will get shorted to R14 in the CARRIER 1 section. This leads to change in carrier frequency i.e. center frequency of Balanced Modulator1 and Band Pass Filter1 causing instability in FDM Transmitter output.

2. Effect Of Switch Fault No. 2 In Band Pass Filter 1 Section

BAND PASS FILTER 1: 7 TO 12 KHz



DESCRIPTION:-

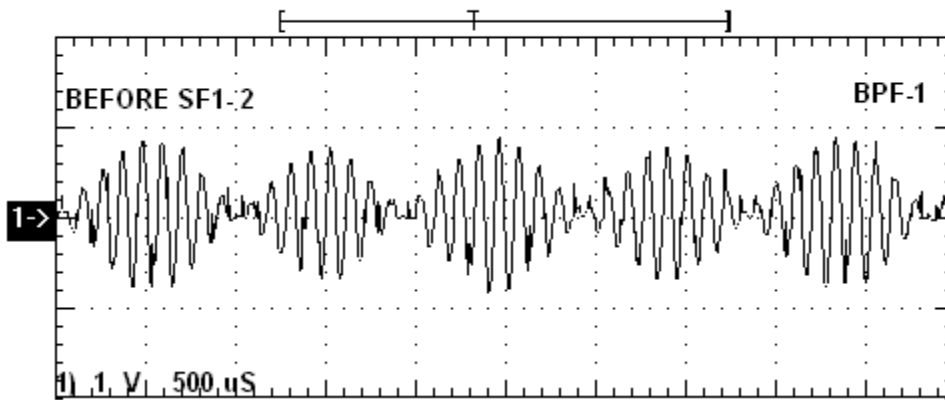
This filter uses only one op-amp. It has two feedback paths, hence the name multiple-feedback filter. The op-amp is used in the inverting mode. The filter is built around op-amp U14 (IC TL082) with pass band frequency range of 7KHz to 12KHz and cut off frequency of 9KHz.

PROCEDURE:-

First keep switch 2 of **SF1** to OFF position.

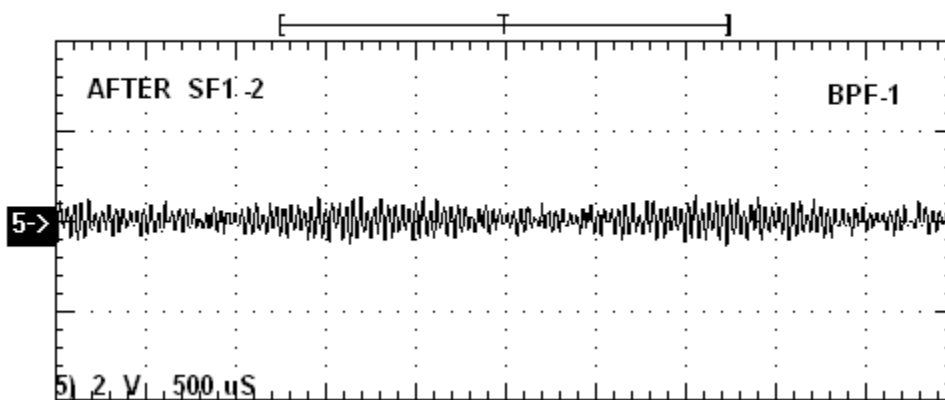
- Connect power supply as per polarity and switch it on.
- Connect the **500Hz, 1Vpp** signal from external function generator to the **MOD IN 1** post of the FDM Transmitter of ACL-FDM.
- Connect the onboard carrier frequency **9KHz, 2Vpp** to the **CARRIER IN 1** post of the FDM Transmitter.
- Keep the **CARRIER NULL** at center position by rotating the pot **P3** as per the procedure for suppressed carrier in previous part, so that the **BM1** is balanced and an AM signal with suppressed carrier is obtained across the output whose amplitude is about 1Vpp.
- Observe the DSB-SC output at the given **BPF1OUT** post of the Band Pass Filter1.

Waveform before switch fault



Now put the switch 2 of **SF1** in the Switch Fault to ON position and observe its effect.

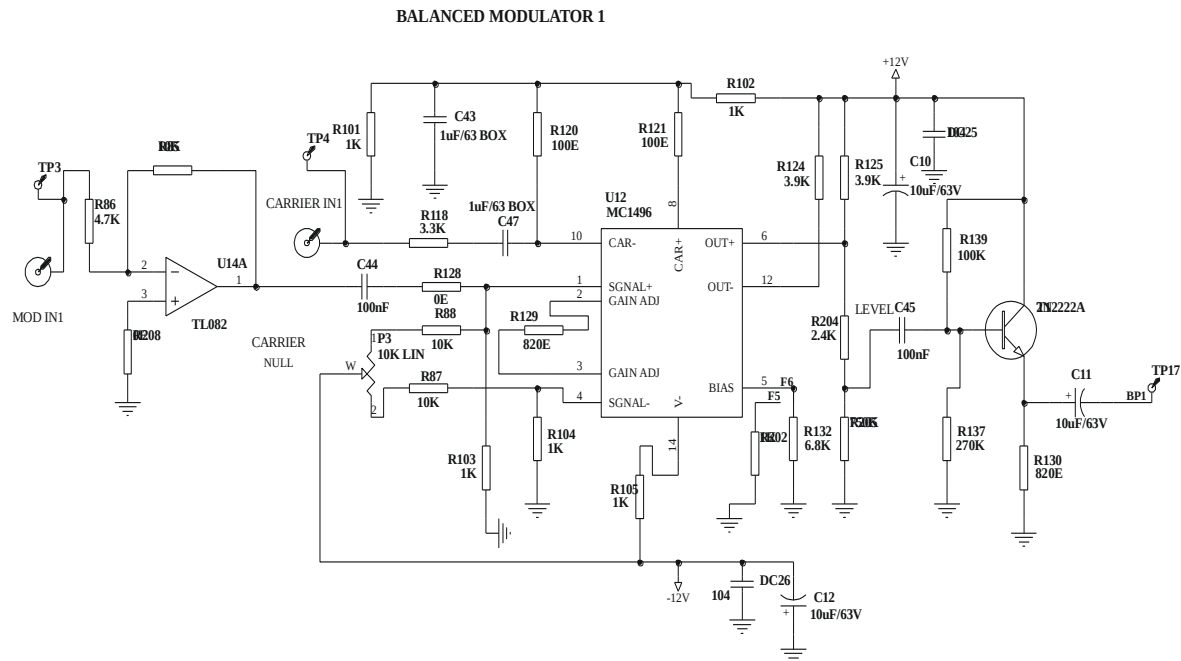
Waveform after switch fault



Effect

When we put the switch 2 of **SF1** in Switch Fault section to ON position, resistor R144 will get shorted to resistor R141 due to which center frequency of **BPF1** gets distorted. This leads to instability in **BPF1** output and further in FDM Transmitter output.

3. Effect Of Switch Fault No. 3 In Balanced Modulator 1 Section



DESCRIPTION:-

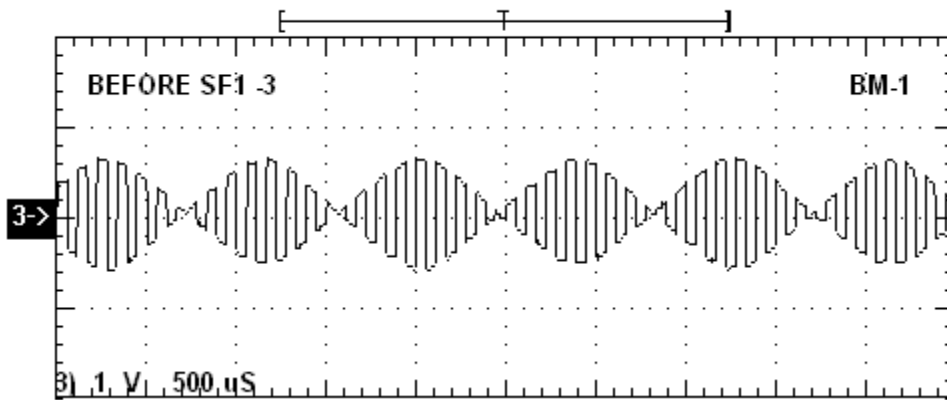
The integrated circuit LM1496 consists in a quadruple differential amplifier, driven by a further differential stage. The output signal consists in a constant, which multiplies the product of the two input signals CARRIER and SIGNAL. The trimmer NULL CARRIER enables the balancing or unbalancing of the circuit: in the first case the output is exactly the product of the input signals, and the circuit operates as balanced modulator; in the second case the output contains a fixed component of the signal CARRIER and the circuit operates as amplitude, which is taken from the transistor emitter.

PROCEDURE:-

First keep switch **3** of **SF1** to OFF position.

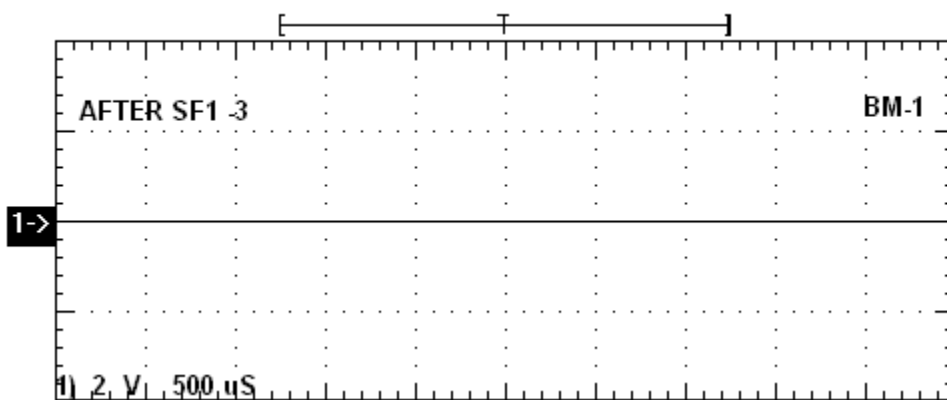
- Connect power supply as per polarity and switch it on.
- Connect the **500Hz, 1Vpp** signal from external function generator to the **MOD IN 1** post of the FDM Transmitter of ACL-FDM.
- Connect the onboard carrier frequency **9KHz, 2Vpp** to the **CARRIER IN 1** post of the FDM Transmitter.
- Keep the **CARRIER NULL** at center position by rotating the pot **P3**, so that the **BM1** is as per the procedure for suppressed carrier in previous part and an AM signal with not suppressed carrier is obtained across the output whose amplitude is about 1Vpp.
- Observe the waveform at the given **BM1** output test point.

Waveform before switch fault



Now put the switch **3** of **SF1** in the Switch Fault to ON position and observe its effect.

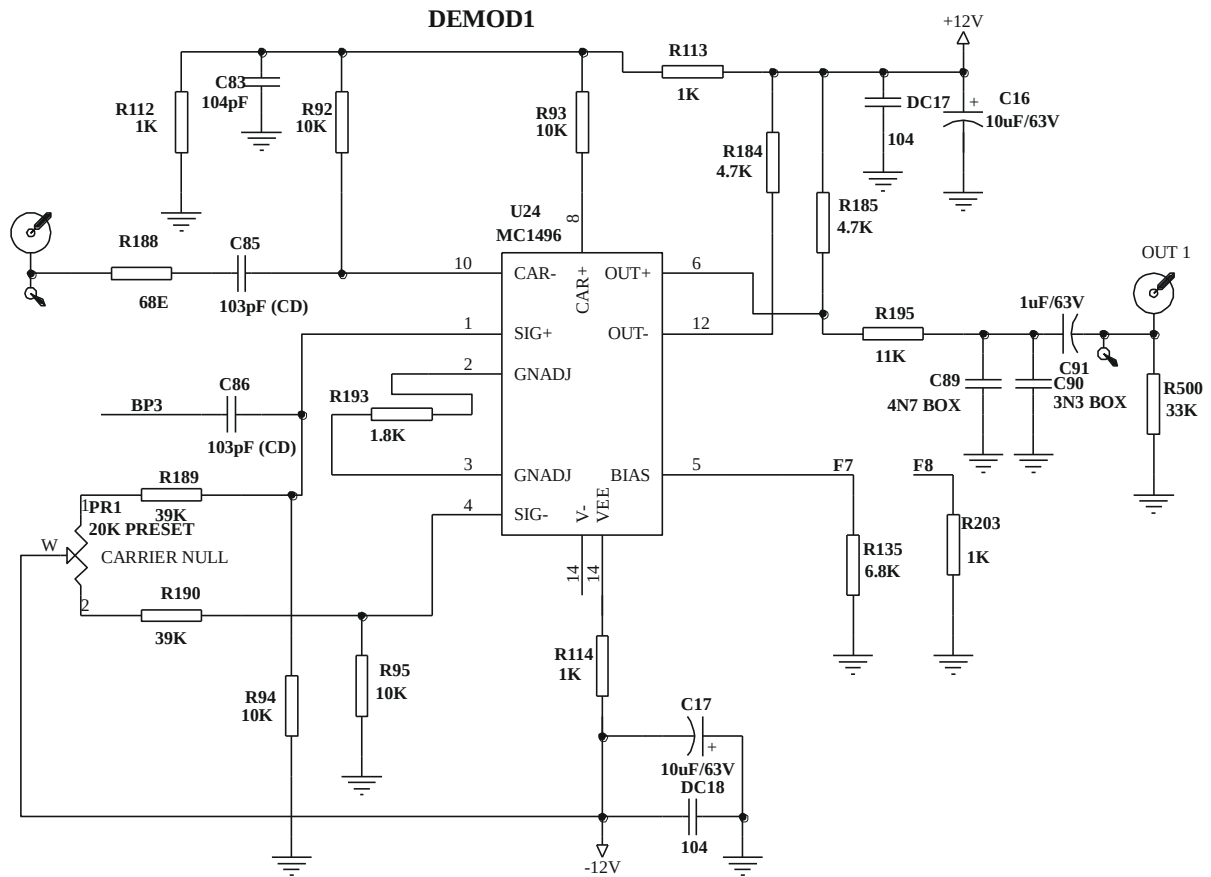
Waveform after switch fault



Action and Effect:

When we put the switch **3** of **SF1** in Switch Fault section to ON position, it will short resistor R202 in parallel with pin no.5 of U12 in the BALANCED MODULATOR 1 section and ground signal. Due to this, biasing voltage of IC U12 gets affected results in disappearance of signal at BM1 output test point.

4. Effect Of Switch Fault No. 4 In Demodulator 1 Section



DESCRIPTION:-

The product detector multiplies the two input signals between them and can be used alone, to demodulate the signals modulated in phase.

This demodulator multiplies signal coming from BPF3 with carrier-1 frequency resulting in demodulated output signal.

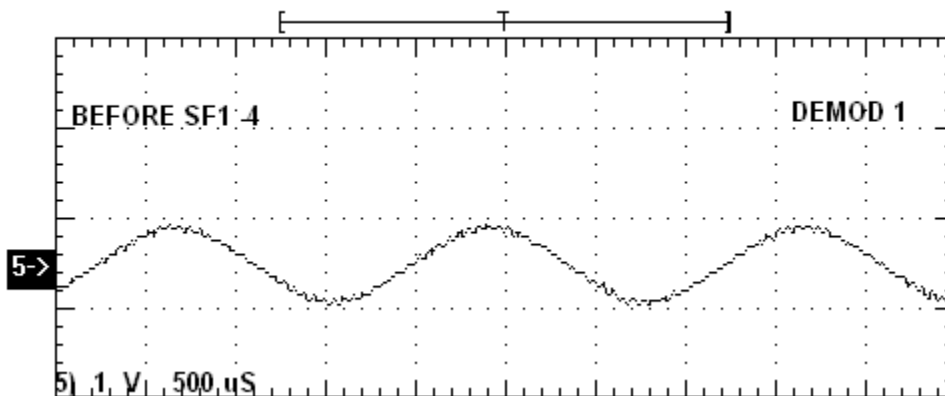
PROCEDURE:-

First keep switch 4 of **SF1** to OFF position.

- Connect power supply as per polarity and switch it on.
- Connect the **500Hz, 1Vpp** Signal from external function generator to the **MOD IN 1** post of ACL-FDM.
- Connect the **1KHz, 1Vpp** Signal from external function generator to the **MOD IN 2** post of ACL-FDM.
- Set carrier frequency **CARRIER 1** to **9KHz** using pot **P1**, set carrier frequency **CARRIER 2** to **20KHz** using pot **P2**.
- Connect the **CARRIER 1** post of the Carrier Generator to the **CARRIER IN1** post of FDM Transmitter and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Transmitter.
- Keep the **CARRIER NULL** at center position by rotating the pot **P3** and **P4**, so that both the modulators are “balanced” and an AM signal with suppressed carrier, as shown in waveform, is obtained across the output whose amplitude is about 1Vpp.

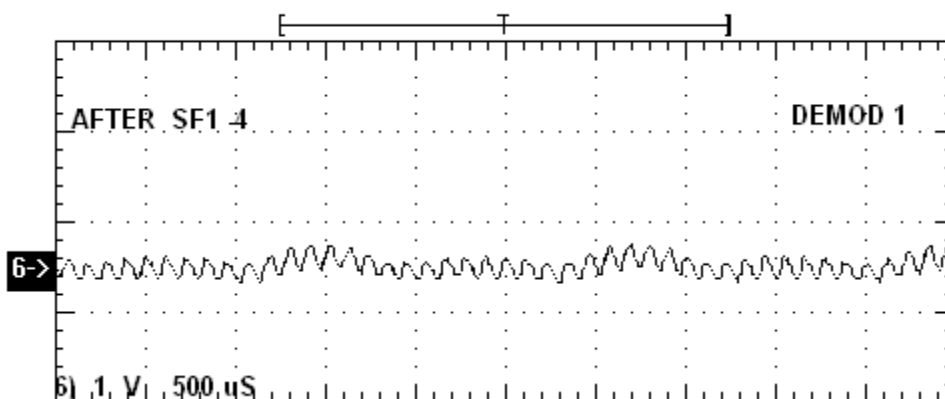
- Connect the FDM signal (which is frequency division multiplexed at frequency **256KHz** of pilot carrier) from the **OUT** post of the FDM Transmitter to the **IN** post of the FDM Receiver.
- Connect the **CARRIER 1** post of the CARRIER GENERATOR to the **CARRIER IN1** post of FDM Receiver and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Receiver.
- Observe the demodulated output at **OUT1** post of the FDM Receiver.

Waveform before switch fault



Now put the switch 4 of **SF1** in the Switch Fault to ON position and observe its effect.

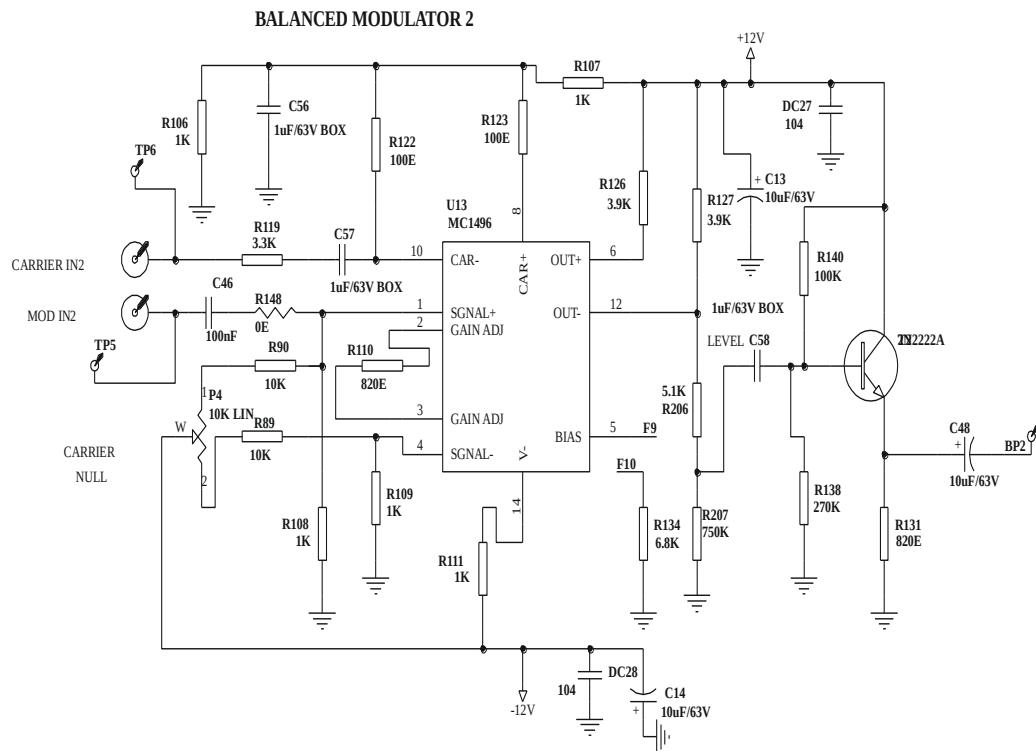
Waveform after switch fault



Action and Effect

When we put the switch 4 of **SF1** in Switch Fault section to ON position, it will short the resistor **R203** in parallel with pin no. 5 of **U24** (Demodulator 1) and ground. This will result in imbalance of bias voltage of IC **U24** and distorted output at **OUT1** post of Demodulator 1 section

5. Effect Of Switch Fault No. 5 In Balanced Modulator 2 Section



DESCRIPTION:-

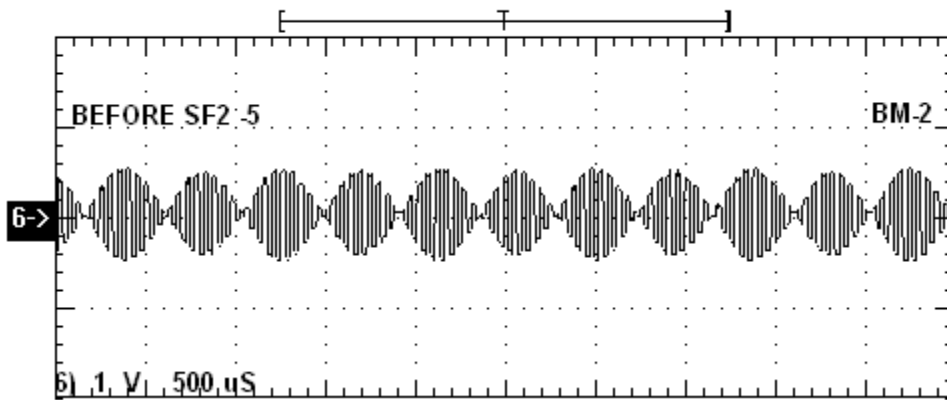
The integrated circuit LM1496 consists in a quadruple differential amplifier, driven by a further differential stage. The output signal consists in a constant, which multiplies the product of the two input signals CARRIER and SIGNAL. The trimmer NULL CARRIER enables the balancing or unbalancing of the circuit: in the first case the output is exactly the product of the input signals, and the circuit operates as balanced modulator; in the second case the output contains a fixed component of the signal CARRIER and the circuit operates as amplitude, which is taken from the transistor emitter.

PROCEDURE:-

First keep switch **5** of **SF2** to OFF position.

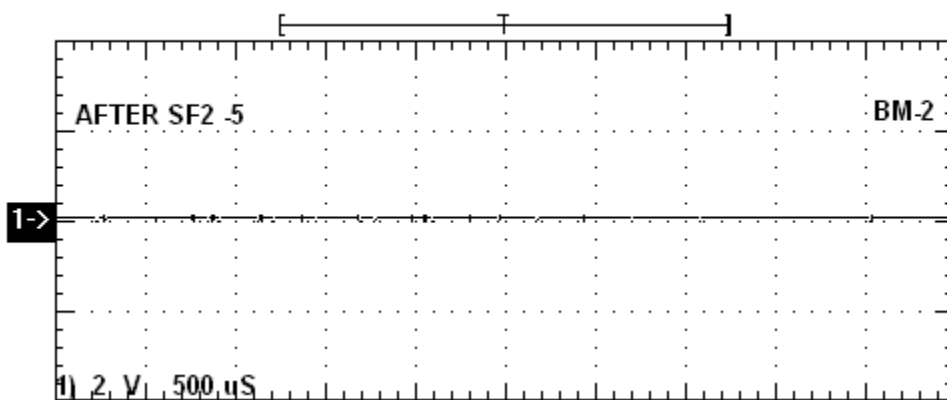
- Connect power supply as per polarity and switch it on.
- Connect the **1KHz, 1Vpp** signal from external function generator to the **MOD IN 2** post of the FDM Transmitter of ACL-FDM.
- Connect the onboard carrier frequency **20KHz, 2Vpp** to the **CARRIER IN 2** post of the FDM Transmitter.
- Keep the **CARRIER NULL** at center position by rotating the pot **P4** as per the procedure for suppressed carrier in previous part, so that the **BM2** is balanced and an AM signal with suppressed carrier is obtained across the output whose amplitude is about 1Vpp.
- Observe the waveform at the given **BM2** output test point.

Waveform before switch fault



Now put the switch **5** of **SF2** in the Switch Fault to ON position and observe its effect.

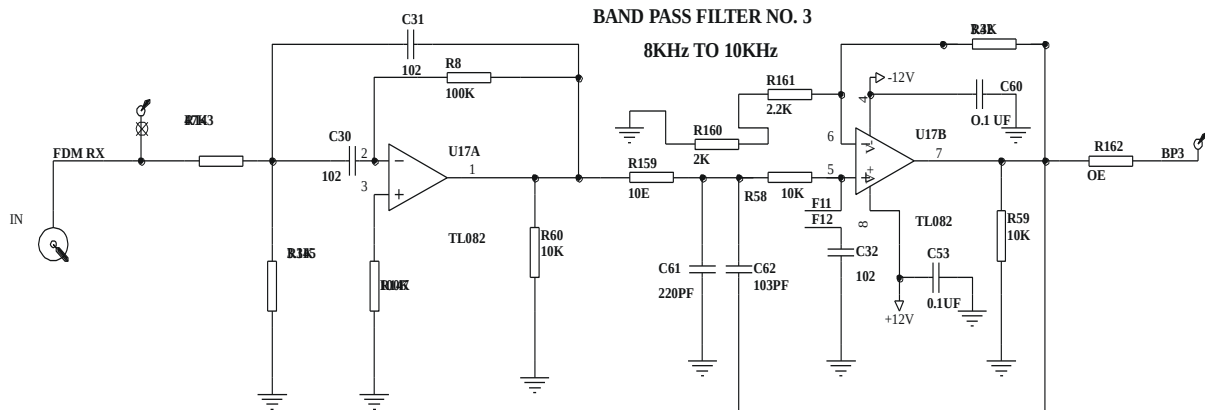
Waveform after switch fault



Action and Effect

When we put the switch **5** of **SF2** in Switch Fault section to ON position, it will open resistor R134 in series from pin no. 5 of U13 in the BALANCED MODULATOR 2 section. It will result in imbalance of biasing voltage for IC U13 due to which output at **BM2** test point will disappear.

6. Effect Of Switch Fault No. 6 In Band Pass Filter 3 Section



DESCRIPTION:-

A wide band pass filter can be formed by simply cascading high pass and low pass sections and is generally the choice for simplicity of design and performance.

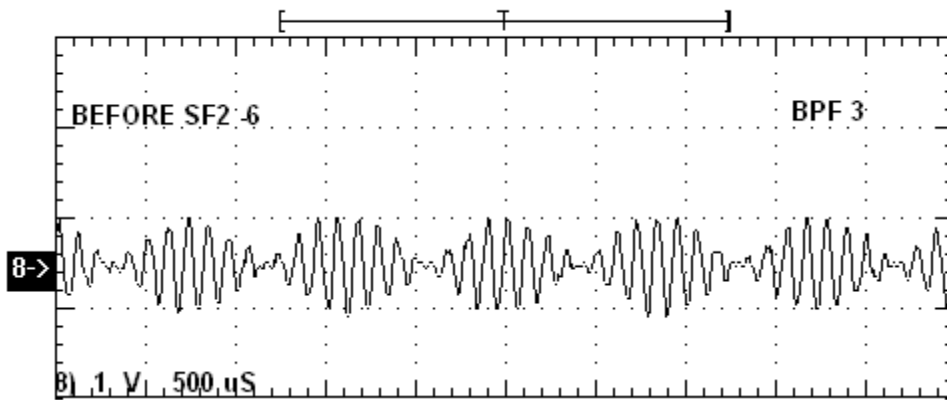
This filter is built around op-amp U17 (IC TL082) with pass band frequency range of 7KHz to 11KHz and cut off frequency of 9KHz.

PROCEDURE:-

First keep switch **6** of **SF2** to OFF position.

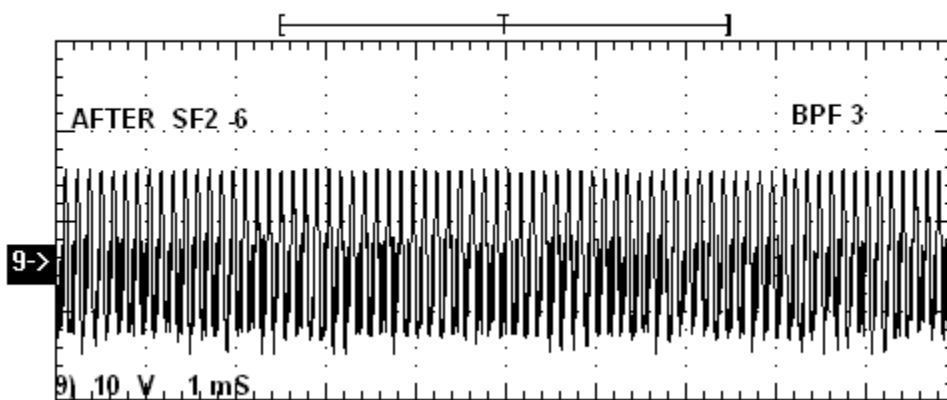
- Connect power supply as per polarity and switch it on.
- Connect the **500Hz, 1Vpp** Signal from external function generator to the **MOD IN 1** post of ACL-FDM.
- Connect the **1KHz, 1Vpp** Signal from external function generator to the **MOD IN 2** post of ACL-FDM.
- Set carrier frequency **CARRIER 1** to **9KHz** using pot **P1**, set carrier frequency **CARRIER 2** to **20KHz** using pot **P2**.
- Connect the **CARRIER 1** post of the Carrier Generator to the **CARRIER IN1** post of FDM Transmitter and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Transmitter.
- Keep the **CARRIER NULL** at center position by rotating the pot **P3** and **P4**, so that both the modulators are “balanced” and an AM signal with suppressed carrier, as shown in waveform, is obtained across the output whose amplitude is about 1Vpp.
- Connect the FDM signal (which is frequency division multiplexed at frequency **256KHz** of pilot carrier) from the **OUT** post of the FDM Transmitter to the **IN** post of the FDM Receiver.
- Connect the **CARRIER 1** post of the CARRIER GENERATOR to the **CARRIER IN1** post of FDM Receiver and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Receiver.
- Observe the Band pass filter output at the given output test point of **BPF3** of the FDM Receiver.

Waveform before switch fault



Now put the switch **6** of **SF2** in the Switch Fault to ON position and observe its effect.

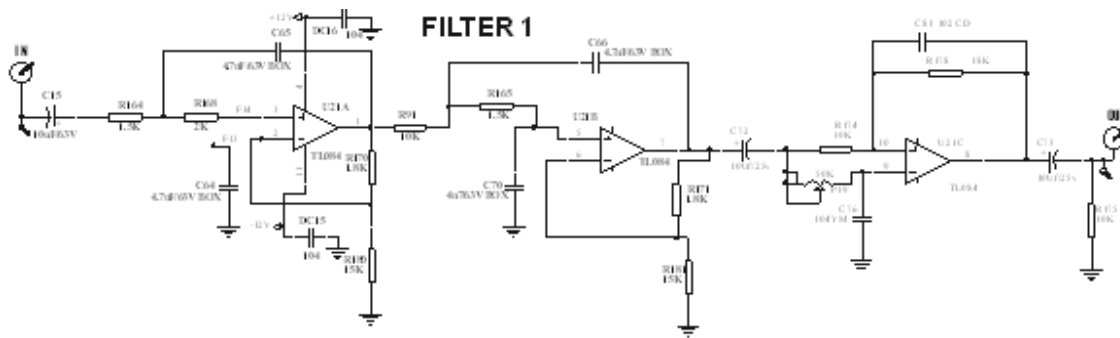
Waveform after switch fault



Action and Effect

When we put the switch **6** of **SF2** in Switch Fault section to ON position, it will open capacitor C32 from pin no. 5 IC U17 of the BPF3. Center frequency of BPF 3 gets distorted, which will result in unwanted signal at the output of **BPF3** section.

7. Effect of switch fault no. 7 in Filter 1 Section



DESCRIPTION:-

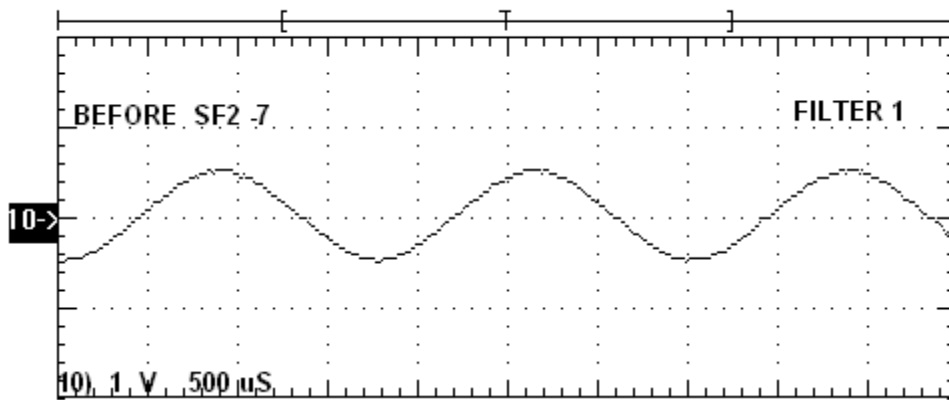
Fourth order low pass butter-worth Filter is formed by U21 (IC TL084) using resistors and capacitors.

PROCEDURE:

First keep switch **7** of **SF2** to OFF position.

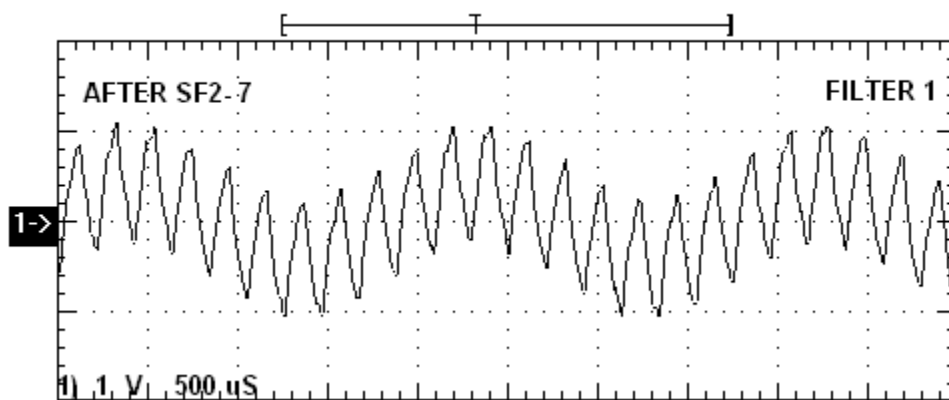
- Connect power supply as per polarity and switch it on.
- Connect the **500Hz,1Vpp** Signal from external function generator to the **MOD IN 1** post of ACL-FDM.
- Connect the **1KHz, 1Vpp** Signal from external function generator to the **MOD IN 2** post of ACL-FDM.
- Set carrier frequency **CARRIER 1** to **9KHz** using pot **P1**, set carrier frequency **CARRIER 2** to **20KHz** using pot **P2**.
- Connect the **CARRIER 1** post of the Carrier Generator to the **CARRIER IN1** post of FDM Transmitter and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Transmitter.
- Keep the **CARRIER NULL** at center position by rotating the pot **P3** and **P4**, so that both the modulators are “balanced” and an AM signal with suppressed carrier, as shown in waveform, is obtained across the output whose amplitude is about 1Vpp.
- Connect the FDM signal (which is frequency division multiplexed at frequency **256KHz** of pilot carrier) from the **OUT** post of the FDM Transmitter to the **IN** post of the FDM Receiver.
- Connect the **CARRIER 1** post of the CARRIER GENERATOR to the **CARRIER IN1** post of FDM Receiver and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Receiver.
- Observe the recovered filtered sine wave at **OUT** post of Filter 1.

Waveform before switch fault



Now put the switch 7 of **SF2** in the Switch Fault to ON position and observe its effect.

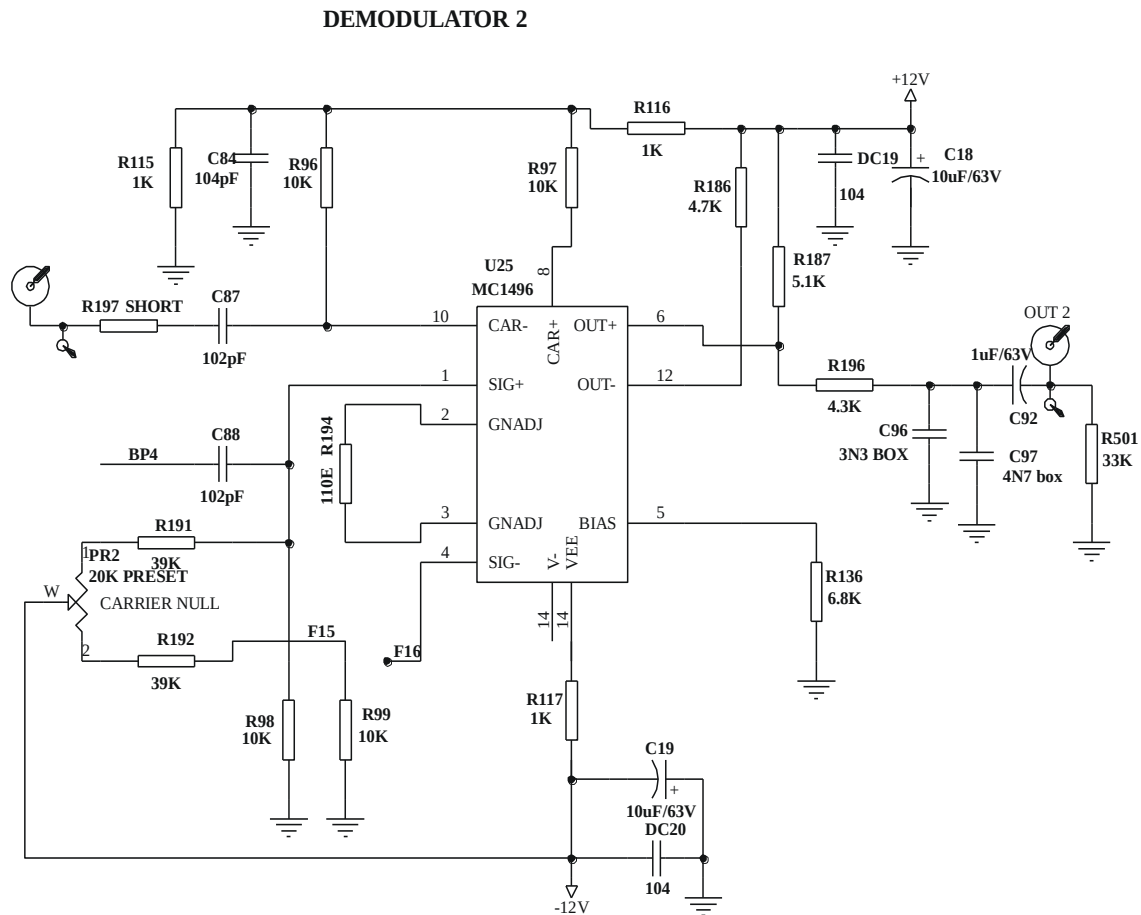
Waveform after switch fault



Action and Effect

When we put the switch 7 of **SF2** in Switch Fault section to ON position, it will open capacitor C64 with pin no. 3 of IC U21 of the FILTER 1 section. It will result in imbalance of filter 1 due to which ripples will be observed at the OUT post of FILTER1.

8. Effect of switch fault no. 8 in Demodulator 2 Section



DESCRIPTION:-

The product detector multiplies the two input signals between them and can be used alone, to demodulate the signals modulated in phase.

This demodulator multiplies signal coming from BPF4 with carrier-2 frequency resulting in demodulated output signal.

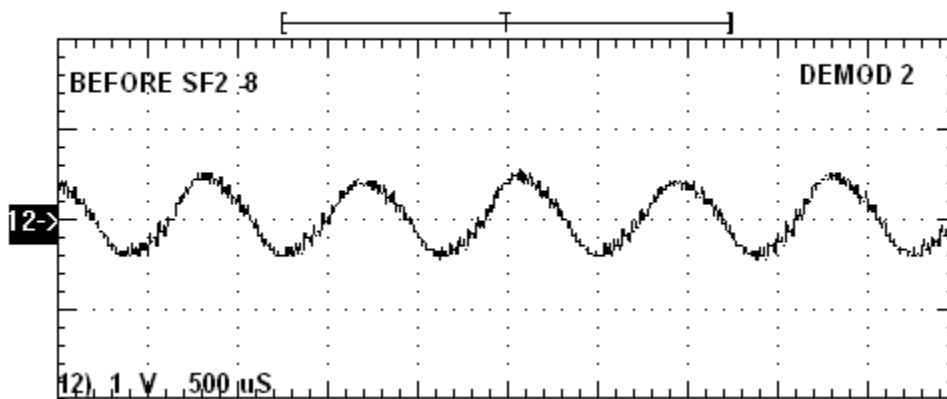
PROCEDURE:-

First keep switch **8** of **SF2** to OFF position.

- Connect power supply as per polarity and switch it on.
- Connect the **500Hz, 1Vpp** Signal from external function generator to the **MOD IN 1** post of ACL-FDM.
- Connect the **1KHz, 1Vpp** Signal from external function generator to the **MOD IN 2** post of ACL-FDM.
- Set carrier frequency **CARRIER 1** to **9KHz** using pot **P1**, set carrier frequency **CARRIER 2** to **20KHz** using pot **P2**.
- Connect the **CARRIER 1** post of the Carrier Generator to the **CARRIER IN1** post of FDM Transmitter and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Transmitter.

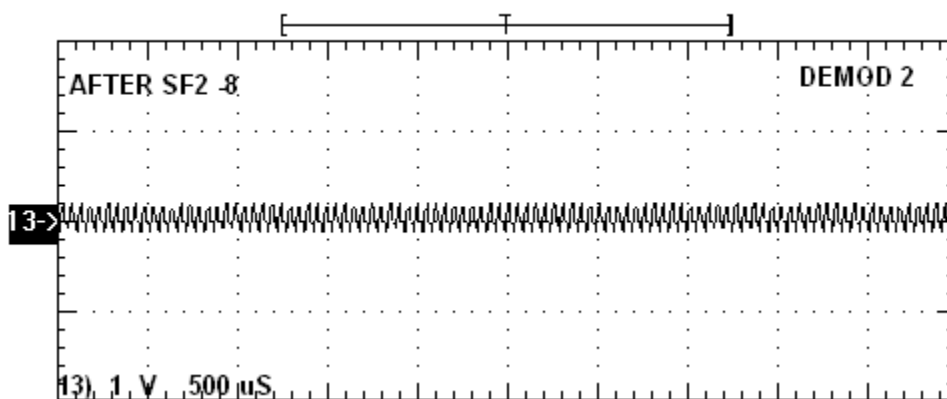
- Keep the **CARRIER NULL** at center position by rotating the pot **P3** and **P4**, so that both the modulators are “balanced” and an AM signal with suppressed carrier, as shown in waveform, is obtained across the output whose amplitude is about 1Vpp.
- Connect the FDM signal (which is frequency division multiplexed at frequency **256KHz** of pilot carrier) from the **OUT** post of the FDM Transmitter to the **IN** post of the FDM Receiver.
- Connect the **CARRIER 1** post of the CARRIER GENERATOR to the **CARRIER IN1** post of FDM Receiver and also connect **CARRIER 2** post of the Carrier Generator to the **CARRIER IN2** post of FDM Receiver.
- Observe the demodulated output at **OUT2** post of Demodulator 2 in FDM Receiver.

Waveform before switch fault



Now put the switch **8** of **SF2** in the Switch Fault to ON position and observe its effect.

Waveform after switch fault



Action and Effect

When we put the switch **8** of **SF2** in Switch Fault section to ON position, it will affect the Demodulator 2 section. Output is not obtained. Due to this, signal connected to pin no. 4 of U25 will be disconnected and it will result in distorted output at output test point of Demodulator 2.